Software Requirements Specification for EyeCanDo: An Ophthalmology Preparation Game for Children

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Table 1: Revision History

Date	Developer(s)	Change
2026/10/12	Mohammad Bilal	Completed sections 1.1, 1.2, 2.1, & contributed to 1.4
2026/10/13	Ahsan Muzammil	Completed sections 2.2, 5, 7, & contributed to section 10, Appendix D, and E
2026/10/13	Mohammad Bilal	Completed section 8, & contributed to section 10, Appendix C, D
2026/10/14	Jackson Beach	Completed Section 1.5 and 2.5
2026/10/15	Jake Finlay	Completed sections 2.3, 6 (including the formal model), 11, 12, & contributed to section 10 and Appendix E
2026/10/15	Ethan McMehen	Completed Sections 1.3, 2.4
2026/10/15	Jackson Beach	Completed Section 4, individual AI usage in Appendix D, and individual reflection in Appendix E
2026/10/16	Ethan McMehen	Completed Sections 3, Appendix A, Contributed to Sections 9, 11, Appendix D, E
2026/10/17	Mohammad Bilal	Made formatting and consistency changes throughout whole document
2026/10/17	Ahsan Muzammil	Completed Appendix B

1 Introduction

This document is the Software Requirements Specification (SRS) for EyeCanDo, a mobile game that prepares children aged 3 to 7 years for paediatric ophthalmology appointments. This document defines the expected behaviour, scope, and requirements of EyeCanDo, establishing a basis for implementation, verification, and validation by describing what the system is expected to accomplish and the conditions under which it will be used.

This document follows the IEEE SRS template provided in the course repository [1], with guidance from ISO/IEC/IEEE 29148 [2] for requirements engineering and ISO/IEC/IEEE 15288 [3] for systems life-cycle management. The team made modifications to the original template to better suit the project's needs. In Section 8 (Non-Functional Requirements), the requirements are organized according to the Volere template provided in the course repository [1]. The AI and Machine Learning Requirements section was also removed, as it is not applicable to the project. A Notations and Conventions section (Section 1.6) was added to explain the identifiers and symbolic constants. A Standards, Laws, and Regulations section (Section 2.6) was added to list the standards, laws, and regulations that apply to the project and the requirements that address them (Table 4).

1.1 Purpose

The purpose of EyeCanDo is to reduce anxiety and uncertainty in children before paediatric ophthalmology appointments by providing accessible, age-appropriate preparation for common eye examination procedures. EyeCanDo aims to improve children's preparedness and willingness to participate while helping parents and healthcare professionals prepare them for appointments. Intended outcomes include reduced anxiety, improved cooperation, sustained engagement across the target age range, more effective preparation at home, and greater clinical efficiency. The application complements, rather than replaces, professional guidance and medical examinations.

1.2 Scope

EyeCanDo is a mobile application intended to help children aged 3 to 7 years become familiar with common paediatric ophthalmology procedures before attending an appointment. The application will present preparation activities as age-appropriate interactive games that allow children to practice actions and learn what they may experience during an eye examination.

The primary users are children preparing for ophthalmology appointments, with parents or guardians supporting their use of the application. Ophthalmologists and child-life specialists are secondary stakeholders who may provide clinical and developmental guidance during development and may use the application as part of a broader preparation process.

The application will provide interactive activities corresponding to common ophthalmology procedures, provide feedback during activities, track completion of preparation activities, and allow users to repeat activities when needed. The application will be designed to support preparation before an appointment, including preparation in a home environment.

The expected benefits are increased familiarity with ophthalmology procedures, reduced uncertainty and anxiety, improved cooperation during examinations, sustained engagement, and potentially shorter or less difficult clinical appointments.

EyeCanDo will not diagnose medical conditions, provide medical advice, replace an ophthalmologist, or determine whether a child requires a particular medical procedure. The application will not replace an actual clinical examination. Medical decisions will remain the responsibility of qualified healthcare professionals.

The initial system will focus on preparation for common ophthalmology procedures and will not attempt to cover every possible procedure or clinical situation. Features identified as stretch goals, such as augmented

reality simulations and personalized preparation based on appointment information, are outside the minimum required scope of the initial system.

1.3 Stakeholders

A stakeholder is anyone who has an interest in the system or can be affected by it. Stakeholders are grouped below using the categories of the IEEE template. Members of the development team are not listed because team members are not stakeholders of their own product.

- **Customer & Project Sponsor**

  A paediatric ophthalmology researcher and clinician who proposed the project, and the paediatric eye clinic that will pilot the product. At the clinic, the customer and project supervisor need children who are prepared for their examinations and evidence of whether the product works in reducing anxiety amongst their patients.

- **End Users**
  - Children aged 3 to 7 years with an upcoming eye appointment.
  - Caregivers (Parents, Guardians, other Adults) who will install the product on their device, set up the game and play the game alongside children.

- **Clinicians**

  Paediatric ophthalmologists and ophthalmic technicians who perform the procedures. Clinicians benefit from calmer and more cooperative patients. They will also tell families which procedures children should practice and review game design as the medical experts in the system.

- **Child-life Specialists**

  They are experts on how children navigate play and the process of medical care. They review the language, imagery and age-appropriate design to advise on the mental, emotional, and social needs of patients [4].

- **Regulatory and Ethics Bodies**

  The research ethics board that approves studies with children, privacy regulators, and the app store operators (Apple and Google), whose policies for children's apps the product must satisfy. Table 4 lists the standards and regulations.

- **Eye Clinics**

  The eye care clinical system benefits when examinations take less time and fewer children need to repeat appointments. Families with limited resources benefit by their children receiving the same preparation for appointments through the free app.

- **Other Paediatric Health-care Professionals**

  Other paediatric health-care professionals, such as family doctors performing check-ups, can reuse the approach as inspiration for their own tools.

- **Society**

  Society benefits when children are better prepared for health-care visits, because calmer children are more likely to receive the care they need and the public health-care system uses its time more effectively. Society also has an interest in children's applications being free, safe, private and inclusive of families from all backgrounds, which EyeCanDo must respect.

- **Administrators**

  EyeCanDo has no server, no accounts and no administration console. There is therefore no administrator role.

1.4 Definitions and Acronyms

Table 2: Definitions and Acronyms

Term	Definition	Location
Accessibility	The design of software so that people with different abilities can use it.	Section 1.1
Android	A mobile operating system developed by Google.	Section 2.5
Apple App Store	Apple's official platform for distributing applications to iOS devices.	Section 2.5
Augmented Reality (AR)	Technology that overlays digital content onto the real world.	Section 1.2
Caregiver Area	The screens meant for adults: the medical notice, the procedure guide, and the settings. Only the settings are behind the parental gate.	Section 6.2
Character Inventory	The collection of cosmetic accessories that a child owns through in-game purchases or direct rewards.	Section 7.1
Child-Life Specialist	A healthcare professional who helps children cope with medical experiences and procedures.	Section 1.2
Cosmetic Accessory	A virtual item, such as a hat or glasses, that changes a character's appearance without affecting gameplay or clinical preparation content.	Section 2.3
Cycloplegic Eye Examination	An eye examination using drops to temporarily relax the eye's focusing muscles and assess refractive error.	Section 2.1
Data Minimization	The practice of collecting and storing only the data necessary for a system's intended purpose.	Section 7.3
Definition (D)	A numbered, boxed statement in the Formal Model that fixes part of the model of the application.	Section 6.1
Eye Drops	Liquid medication applied to the eyes, including drops used to dilate pupils or temporarily relax focusing muscles.	Section 2.1
Eye Examination	A clinical assessment of eye health and vision.	Section 1.2
First Completion Reward	The journey points, and any coins or accessory, given when a level is finished for the first time.	Section 5.3
Exercise	An interactive task inside a level that the child performs, such as moving a virtual patient or answering a question. Narration and animations are not exercises.	Section 6.2

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Table 2 – continued from previous page

Term	Definition	Location
Fit Criterion	A measurable condition used to determine whether a requirement has been satisfied.	Section 8.1
Functional Requirement (FR)	A specification of a capability or behaviour the application must provide.	Section 5.1
Game Progress	The recorded completion status of a preparation game: Not Started, In Progress, or Completed.	Section 5.2
Home Control	The control on every screen of a level that leaves the level and returns to the journey map.	Section 6.2
Home Screen	The first screen after launch and setup. It leads to the journey map, character selection, the procedure guide, and the settings.	Section 5.1
In-Game Coins	A virtual currency earned by finishing levels and spent on cosmetic accessories. It cannot be bought with real money.	Section 5
iOS	Apple's mobile operating system for iPhone devices.	Section 2.5
Journey-Based Progression	A game structure in which children explore ophthalmology preparation games presented as levels within a journey. All levels remain accessible regardless of previous completion.	Section 7
Journey Map	The screen that shows every level of the journey and its status, and from which the child starts a level.	Section 6.2
Journey Level	A stage within the preparation journey corresponding to a procedure game. Each level may contain multiple interactive exercises or rounds.	Section 5.10
Journey Points	Points earned through participation in preparation games. They contribute to the child's overall score and are not reduced when in-game currency is spent.	Section 5
Likely Change (LC)	A change to the requirements or assumptions that is expected during the project or after release.	Section 11
Low-Spec Device	A reference device released at least `LOW_SPEC_AGE_YEARS` years before the test date that still supports the minimum operating system version (NFR24).	Section 8.5
Non-Functional Requirement (NFR)	A specification of the application's quality attributes, performance expectations, or constraints.	Section 1
Ophthalmologist	A physician specializing in eye health, vision, and eye conditions.	Section 1.2
Paediatric Ophthalmology	The medical specialty focused on children's eye health and conditions.	Section 1

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Table 2 – continued from previous page

Term	Definition	Location
Parental Gate	A task in front of the settings that requires reading a written instruction, so that an adult can complete it and a pre-reading child is unlikely to.	Section 6.2
Personal Scoreboard	A screen that shows one child's own level statuses, points, and coins, without comparison to other children.	Section 5
Personally Identifiable Information (PII)	Information that can identify an individual directly or indirectly.	Section 7.3
Privacy	The protection of personal information and control over how it is collected, used, stored, and disclosed.	Section 2.3
Procedure Game	An interactive game representing one ophthalmology procedure, such as vision testing, eye-drop administration, or slit-lamp examination.	Section 2.3
Question	An exercise in which the child chooses from displayed answers, at least one of which is correct.	Section 6.2
Reference Device	A phone or tablet in the test set of the VnV Plan. The set includes at least one iOS and one Android device, and at least one phone and one tablet.	Section 8.5
Requirements Traceability Matrix (RTM)	A table linking stakeholder requirements, use cases, and functional and non-functional requirements to support consistency and completeness.	Section 9
Slit-Lamp Examination	An examination in which the clinician uses a microscope with a bright, narrow beam of light to look at the front of the eye while the patient rests their chin and forehead on the device.	Section 2.1
Software Requirements Specification (SRS)	A document defining a software system's requirements, scope, behaviour, and constraints.	Section 1
Stakeholder	An individual or organization that affects, is affected by, or has an interest in the system.	Section 1.2
Stakeholder Requirement (STR)	A statement of a stakeholder's needs or expectations.	Section 3
Theorem (T)	A numbered property of the Formal Model that the functional requirements depend on.	Section 6.1
Touch Target	The screen area that a user can tap to activate an interactive element.	Section 8.2
Unlikely Change (ULC)	An assumption that the design can rely on because it is not expected to change.	Section 12
Use Case (UC)	A description of how an actor interacts with the application to achieve a goal.	Section 5
Verification and Validation (VnV) Plan	A document describing how the system will be checked against its specified requirements and user needs.	Section 8.5

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Table 2 – continued from previous page

Term	Definition	Location
Web Content Accessibility Guidelines (WCAG)	Guidelines published by the World Wide Web Consortium (W3C) for making digital content more accessible.	Section 8.2

1.5 Characteristics of Intended Reader

The intended readers are the individuals who will read, review and maintain this document. Their characteristics are:

- An understanding of requirements engineering, at least at the level of an upper-year undergraduate course in a software engineering/computer science program. This is so the reader can understand the meaning behind the structure (e.g., Personas, Scenarios, etc.) and terminology (e.g., Functional vs. Non-Functional Requirements) used.
- An understanding of discrete mathematics, at least at the level of an undergraduate course in a software engineering/computer science program. This is so the reader can understand the notation used in formal specifications of the system's behaviour.
- Familiarity with touchscreens and mobile applications/games, at a general level expected of someone who has interacted with such applications as a consumer.
- Familiarity with the problem the system is attempting to address, at a level which can be achieved from reading the Problem Statement document.

Knowledge of clinical terms and processes is not a prerequisite for the intended reader. Such terminology and processes are defined in Section 1.4 and Section 1.2, respectively.

1.6 Notations and Conventions

Identifiers

Every constraint, requirement, use case, change, and formal definition has a unique identifier. Each identifier has a prefix that says what it is, followed by a number that is assigned in order of appearance, as in Table 3. Every use of an identifier outside its own definition is a link to that definition, so the reader can move in both directions between an item and the places that use it.

Identifiers written with a hyphen, such as SR-11 (a safety and security requirement) and H2-9 (a hazard), belong to the Hazard Analysis. They link to the published Hazard Analysis and not to this document.

Symbolic Constants

A value that several requirements depend on is given a symbolic constant. A constant is written in monospaced capital letters with underscores, such as `TOUCH_TARGET_MIN`. Its name ends with the kind of bound, `MIN` or `MAX`, and, for a quantity that has a unit, the unit, such as `INSTALL_SIZE_MAX_MB`. Each use of a constant is a link to its value in Table 20.

Table 3: Identifier Prefixes

Prefix	Item	Where Defined
C	Constraint	Section 2.5
STR	Stakeholder requirement	Section 3
UC	Use case	Section 5
FR	Functional requirement	Section 6.2
NFR	Non-functional requirement	Section 8
D, T	Definition and theorem of the formal model	Section 6.1
LC	Likely change	Section 11
ULC	Unlikely change	Section 12
ER	Elicitation source of the stakeholder requirements	Appendix 13.1.1
P1, P2, P3	Requirement priority	Appendix 13.2
E	Equation of the formal model	Section 6.1

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

2.1 Problem Statement

Paediatric ophthalmology appointments can be stressful and unfamiliar for children. Procedures such as vision testing, eye drops, and slit-lamp examinations may cause anxiety, particularly when children do not know what to expect. This anxiety can make it difficult to cooperate during examinations, potentially affecting the quality and completeness of clinical assessments. For example, a study of 298 children undergoing cycloplegic eye examinations found that 26% experienced significant distress before receiving eye drops, while uncooperative children were also more anxious and required more time to complete the examination [5].

Children and their families therefore need adequate preparation for common ophthalmology procedures before attending an appointment. Without appropriate preparation, unfamiliar clinical experiences may contribute to increased anxiety, reduced cooperation, and longer or more difficult appointments. This project addresses the need for an accessible, child-friendly way to help children become more familiar and comfortable with the paediatric ophthalmology experience.

2.2 Product Vision

EyeCanDo is an interactive mobile application designed to help children aged 3–7 feel more comfortable and prepared for paediatric ophthalmology appointments. Through age-appropriate games and activities, the application introduces children to common procedures, including vision testing, eye-drop administration, and slit-lamp examinations. By allowing children and their families to become familiar with these procedures before their appointments, EyeCanDo aims to reduce anxiety, improve cooperation during examinations, and support a more efficient clinical experience.

The primary goals of EyeCanDo are:

- **Reduce Children's Anxiety:** Help children understand what to expect during common ophthal-

mology procedures, reducing anxiety associated with unfamiliar clinical experiences.

- **Improve Children's Cooperation During Examinations:** Prepare children to participate more cooperatively in clinical examinations, reducing interruptions and repeated attempts during procedures.
- **Maintain Engagement Across Age Groups:** Provide engaging and age-appropriate activities that encourage children across the target age range to actively participate in their preparation.
- **Enable Effective At-Home Preparation:** Allow children and their families to independently prepare for upcoming ophthalmology appointments through accessible and understandable activities.
- **Improve Clinical Efficiency:** Support more efficient ophthalmology examinations by improving children's preparedness and reducing difficulties associated with completing clinical procedures.

These goals establish the intended outcomes of EyeCanDo and provide the foundation for the functional and non-functional requirements defined in this document.

2.3 System Context

Figure 1 shows EyeCanDo and the entities around it. EyeCanDo runs entirely on one phone or tablet. It has no accounts, servers or network connection while it runs (C6). Every input and output therefore passes between the application and either its users or the device. Content reaches the device only through the app store installation package. The clinical and child-life reviewers approve that content before release.

The game engine and other third-party libraries are packaged inside the application. They are part of EyeCanDo, not external entities, so the figure does not show them. The engine is also not named, because C4 allows it to change.

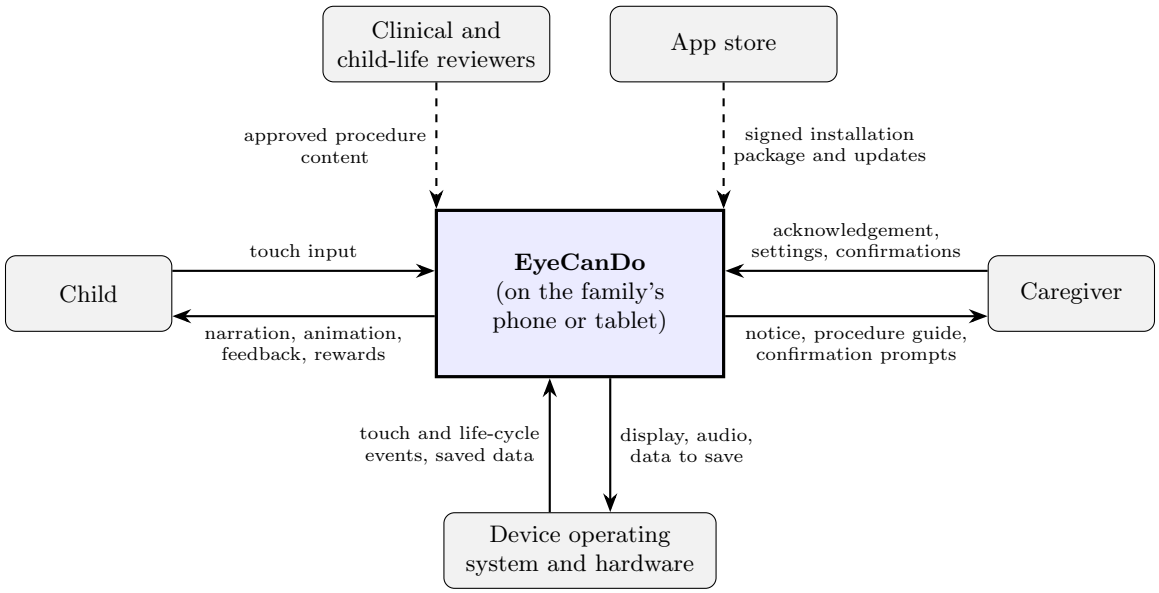


Figure 1: System context of EyeCanDo. Solid arrows are data flows while the application runs. Dashed arrows happen before the application runs, at release or installation.

The touch input and audio-visual output of the child and caregiver physically pass through the device. The figure shows the two users separately because they provide different inputs and need different outputs. The responsibilities of each entity are listed below in terms of its inputs and outputs.

- **Child:**

  - Provides touch input to choose a character, select levels, play the procedure games and spend coins on cosmetic accessories.
  - Receives the preparation content, feedback and rewards through pictures, animation and narration, without needing to read.

- **Caregiver:**
  - Installs EyeCanDo and acknowledges the medical notice before first use (FR2).
  - Enters local preferences, such as the child's optional nickname and age group (FR40, FR41).
  - Confirms actions that cannot be undone, such as resetting saved data (FR42).
  - Reads the procedure guide and the notice.
  - Supervises younger children as needed.
- **Device operating system and hardware:**
  - Delivers touch events and life-cycle events, such as the application being sent to the background, closed or resumed.
  - Displays images and plays audio at the volume the caregiver sets.
  - Stores and returns the application's saved data in storage that only the application can read (NFR26).
- **Clinical and child-life reviewers:**
  - Approve the procedure content before each release. This includes the steps, equipment and sensations shown.
  - Confirm that the content is suitable for children aged 3 to 7 (NFR2).
- **App store:**
  - Delivers the signed installation package and later updates (NFR27, C8). The package contains all procedure content.
- **EyeCanDo:**
  - Presents the approved procedure games and caregiver information without a network connection (C5).
  - Accepts only the touch inputs expected on the current screen.
  - Ignores repeated or accidental touches that would skip procedure information.
  - Records progress, rewards, character choices and preferences (Section 6.1).
  - Stores no identifying information or clinical records (C7).
  - Checks saved data when it is loaded. If the data is missing or invalid, the application recovers without blocking access to any level (FR27).
  - Stops all audio and animation of a level when the child exits it.
  - Keeps the child's progress when the application is interrupted (FR25, FR26).
  - Never transmits data off the device (NFR28).

Context of use: EyeCanDo is used for health education. Families use it at home before an appointment or in a clinic waiting room. It is not mission-critical. It is also not safety-critical in the sense of controlling equipment or informing clinical decisions. It makes no diagnostic or treatment claim (NFR32).

However, its users are young children in a medical setting. Some failures can increase a child's anxiety or reduce cooperation at the appointment, which is the opposite of the application's purpose. Examples are inaccurate or frightening content, hidden sensations, punishing feedback, privacy failures, and a caregiver treating the game as medical advice. The Hazard Analysis document analyses these hazards. The resulting safety and security requirements are included in this SRS. Requirements that address these hazards receive a fuller rationale than other requirements.

2.4 User Characteristics

The users are different from the intended readers of this document (Section 1.5). The characteristics below are used to set the non-functional usability requirements in Section 8.2. The characteristics are also illustrated by the personas in Section 4.

Children

- Aged 3 to 7 years. Many cannot read, and older children may read only simple words. All instructions must therefore be spoken or narrated and shown with pictures or visual guides.
- Actively developing fine motor control. Children in this age range tap imprecisely, often with several fingers, and may tap repeatedly or not at all.
- Have short attention spans, which grow with age.
- May have reduced vision at the time of use because they are eye patients. Therefore, some wear patches or glasses. Additionally, in the waiting room their pupils may be dilated, making the screen look blurry and bright. Some have colour vision deficiencies.
- May already be anxious about the appointment and respond strongly to praise and to perceived failure.
- May be highly dependent on their caregiver. This can make the child shy with other adults, reluctant to take part in the examinations at the appointment, and reluctant to try things on their own, such as using EyeCanDo in the waiting room.
- The child may not have used technology before. The game may then have a steep learning curve, which adds its own anxiety or nervousness.

Caregivers

- Adults with varied education, technical experience and English proficiency.
- Adults with varied accessibility needs, like their own vision impairment (notable in an eye clinic).
- They can operate a smartphone app store and read simple English instructions.
- Often stressed or short of time and may use EyeCanDo in a busy waiting room alongside their child.
- Have no clinical training.
- May be tired after a full day of work, other activities, or general external factors (e.g. family illness, other child responsibilities) affecting level of patience.
- They can legally consent to examinations, share medical history, or approve data collection on behalf of their children.

Clinicians

- Eye-care professionals with intensive clinical procedure knowledge.
- Eye-care professionals who use and administer specialized equipment such as the slit-lamp, Optical Coherence Tomography (OCT) scanners for eye photos, eye-drop medications, and other vision test equipment.
- Have relatively little time per patient as appointment slots are often full, so introducing EyeCanDo to a patient must take at most `CLINICIAN_FIND_MAX_MIN` minute (NFR6).
- Have large schedules packed with patient appointments or other healthcare responsibilities, so training in EyeCanDo must take at most `CLINICIAN_TRAINING_MAX_MIN` minutes (NFR6).
- Experience frequent interruptions from restless children and have to face multiple caregiver questions.
- Must abide by standards and ethics set by regulatory bodies regarding patient privacy, security, and personal information.

Child-life Specialists

- Certified and trained in child development, play, and medical care navigation.
- Have clinical experience and clinical training.
- Can share the child's coping plan with clinicians so that everyone uses consistent language and approaches with the child.
- They can operate a smartphone app store and read English instructions with professional fluency.
- Provide information, support, and guidance to parents and family members about examinations and medical procedures [4].
- Collaborate with clinicians to coordinate and manage care [4].
- Must abide by standards and ethics set by regulatory bodies regarding patient privacy, security, and personal information.

2.5 System Constraints

Each constraint lists the stakeholder requirements (Section 3) that it supports.

C1: EyeCanDo shall operate on consumer smartphones and tablets running iOS or Android.

Rationale: Patient families and clinics may use mobile devices running either operating system. Supporting both enables preparation at home and in clinic without assuming a particular platform.

Stakeholder Requirements: STR10.

Refined by: NFR24, which gives the fit criterion.

C2: Third-party assets and libraries included in EyeCanDo shall have licenses allowing their use and redistribution within the application. Required copyright notices, licenses, and attributions shall be retained.

Rationale: The application's source code is licensed under the MIT license, while third-party materials have their own licensing. These conditions must be satisfied when the application is released.

Stakeholder Requirements: STR16.

Refined by: NFR33, which gives the fit criterion.

C3: Third-party materials present in the EyeCanDo GitHub repository shall have licenses allowing redistribution in the form committed to the repository.

Rationale: Permission to distribute an asset in an application does not necessarily mean redistribution of its raw files in the project repository is permitted. Repository contents must satisfy the applicable licensing conditions (separately from the compiled application).

Stakeholder Requirements: STR16.

C4: EyeCanDo shall be built using a game engine with native support for cross-platform deployment to iOS and Android.

Rationale: One code base that deploys natively to both platforms satisfies C1 with a single implementation. The team of five has two academic terms to build the application, which is not enough time to build and test two separate native applications. A game engine also provides the animation, audio, and touch handling that the levels need. The team has currently selected the Unity game engine, but the selected engine may change provided that it satisfies this constraint.

Stakeholder Requirements: STR10.

C5: EyeCanDo shall provide all included preparation modules and their required assets without internet connection after installation.

Rationale: Families must be able to prepare at home and in clinic, where connectivity may be poor or unavailable.

Stakeholder Requirements: STR4.

Refined by: NFR19, which gives the fit criterion.

C6: EyeCanDo shall operate as a standalone local application and shall not initiate network communications during operation, including through included third-party libraries.

Rationale: Preparation activities do not require external services or data transit. Prohibiting network communications also prevents included libraries from transmitting usage data. Downloading and updating the application through its distribution channels are outside normal application operation (i.e., a patient progressing through the application's preparation modules).

Stakeholder Requirements: STR13.

Refined by: NFR28, which gives the fit criterion.

C7: EyeCanDo shall not request, collect, or store identifying information or clinical records, and shall not integrate with clinic information systems.

Rationale: Preparation activities do not require patient records. Limiting application data to local gameplay, like character selection and scores, supports preparation without requiring identifying information.

Stakeholder Requirements: STR13.

Refined by: NFR29, which gives the fit criterion.

C8: The iOS and Android releases of EyeCanDo shall comply with the Apple App Store and Google Play submission requirements, respectively, including applicable policies for applications meant for children,

at time of submission.

Rationale: These marketplaces are the planned distribution channels through which families and clinics will install the application. Compliance with their submission requirements is necessary for distribution to the target users. The policies are listed in Table 4.

Stakeholder Requirements: STR15, STR16.

Refined by: NFR34, which gives the fit criterion.

2.6 Standards, Laws, and Regulations

Table 4 lists the standards, laws, and regulations that apply to EyeCanDo, or that were considered, and the requirements that address them.

Table 4: Applicable Standards, Laws, and Regulations

Source	Relevance	Addressed By
ISO/IEC/IEEE 29148 [2] and ISO/IEC/IEEE 15288 [3]	Guidance for requirements engineering and for systems life-cycle management.	The structure of this document
Web Content Accessibility Guidelines 2.2, Level AA [6]	Accessibility benchmark for children with low vision, hearing, or motor differences. Success criterion 2.3.1 limits flashing content.	NFR9, NFR16
Accessibility for Ontarians with Disabilities Act, 2005 [7]	Accessibility duty in Ontario.	NFR9
Health Canada guidance on software as a medical device [8]	Software that claims to diagnose, screen, treat, or monitor a condition may be regulated as a medical device. The application is an educational aid and makes no such claim.	NFR32, FR1
Personal Information Protection and Electronic Documents Act (PIPEDA)	Canadian privacy law for personal information. The application collects none.	C7, NFR28, NFR29
Personal Health Information Protection Act, 2004 (PHIPA)	Ontario law for personal health information held by clinics. The application holds none and does not connect to clinic systems.	C7
Apple App Review Guidelines [9]	Rules for children's applications (Kids Category) that must be met for distribution on the Apple App Store.	C8, NFR34
Google Play Developer Program Policies [10]	Policies for children's applications (Families policy) that must be met for distribution on Google Play.	C8, NFR34
MIT licence and the licences of third-party assets and libraries	Conditions for using and redistributing the application's code and assets.	C2, C3, NFR33

3 Stakeholder Requirements

The stakeholder requirements state, in the stakeholders' own terms and without reference to technology, what EyeCanDo must achieve. Each requirement is traced to its rationale from the elicitation sources in Appendix 13.1.1 (Table 10). Each requirement also has a *Refined by* line that lists the use cases, non-functional requirements, and system constraints that satisfy it. Each of those lists the stakeholder requirements it satisfies, so the trace can be followed in both directions.

STR1: Children shall be able to learn what happens during each common paediatric eye clinic examination before their appointment.

Rationale: ER1, ER7

Refined by: UC3, UC4, UC5, UC6, UC7, NFR8, NFR14, NFR22.

STR2: The preparation shall calm children and reduce anxiety.

Rationale: ER1

Refined by: UC5, UC6, UC7, UC8, NFR2, NFR4, NFR12, NFR13, NFR15, NFR17, NFR25.

STR3: The product shall be engaging and tailored for children aged 3 to 7.

Rationale: ER1, ER2

Refined by: UC3, UC5, UC6, UC7, UC9, UC10, UC11, UC12, UC13, NFR1, NFR3, NFR5, NFR8, NFR11, NFR13.

STR4: Families shall be able to prepare for the examinations on their own at home and while at the clinic waiting room.

Rationale: ER1

Refined by: UC1, UC2, UC3, UC5, UC6, UC7, UC15, NFR5, NFR19, NFR20, NFR21, C5.

STR5: Preparation content shown to children shall match what actually happens in the clinic, as confirmed by clinicians.

Rationale: ER1

Refined by: UC4, UC5, UC6, UC7, NFR22, NFR27.

STR6: Children shall be told truthfully about unpleasant sensations, such as drops that burn or bright lights in their eyes.

Rationale: ER1, ER5

Refined by: UC4, UC5, UC6, UC7, NFR2, NFR22.

STR7: Families shall be able to access all preparation with focus on the procedures planned for the child's own visit, as chosen by the clinician or family.

Rationale: ER1, ER3, ER8

Refined by: UC3, UC4, UC5, UC6, UC7, UC8, NFR6.

STR8: The product preparation shall include a journey with different levels for children to earn and see their rewards for practicing.

Rationale: ER1, ER8

Refined by: UC3, UC9, UC10, UC11, UC12, NFR7.

STR9: Caregivers, clinicians and the evaluation study shall be able to see which preparation activities a child has completed.

Rationale: ER3

Refined by: UC3, UC8, UC9, UC10.

STR10: The product shall work on the phones and tablets that families already own.

Rationale: ER3

Refined by: UC1, UC15, NFR11, NFR21, NFR24, C1, C4.

STR11: It shall be possible to add more clinic procedures to the existing interface over time.

Rationale: ER5

Refined by: UC3, UC4, NFR23.

STR12: The product shall include the ability to play as a character customized by the child.

Rationale: ER2, ER8

Refined by: UC2, UC11, UC12, NFR31.

STR13: The product shall not hold personal health information of families and such information shall stay in the clinic's existing secure systems.

Rationale: ER6

Refined by: UC1, UC13, UC14, UC15, NFR26, NFR28, NFR29, NFR30, C6, C7.

STR14: Families shall understand that the product is a preparation aid and does not give medical advice.

Rationale: ER4

Refined by: UC1, UC4, NFR32.

STR15: The product shall include safe, age-appropriate content free of advertising, real-money purchases, and frightening content.

Rationale: ER2, ER3

Refined by: UC5, UC6, UC7, UC11, UC15, NFR2, NFR15, NFR16, NFR18, NFR34, C8.

STR16: The product shall comply with the rules and ethics for children's apps set by app store guidelines and regulatory ethics bodies, including avoiding designs that encourage compulsive use.

Rationale: ER3

Refined by: UC3, UC9, UC10, UC11, UC13, UC14, NFR7, NFR27, NFR33, NFR34, C2, C3, C8.

STR17: The product shall be accessible to children with visual impairments, including those with dilated pupils, eye patches, or colour vision deficiency, as well as children who are deaf or hard of hearing.

Rationale: ER3, ER5

Refined by: UC5, UC6, UC7, UC13, NFR3, NFR9, NFR10.

STR18: Children with pre-reader ability shall be able to use the product.

Rationale: ER2, ER3

Refined by: UC2, UC3, UC5, UC6, UC7, NFR1, NFR5, NFR8.

4 User Personas

These are fictitious personas drafted from project documentation. Their individual characteristics are assumptions that will be validated through further stakeholder interaction.

4.1 Persona 1: Four-Year-Old Paediatric Patient

- **Name:** Noah.
- **Description:** Noah is a four-year-old boy who has been referred to an ophthalmologist after his mother noticed that he appeared to have difficulty seeing objects on screens and paper. He often sits close to the TV or brings the family's tablet close to his face while playing. He cannot read independently, but enjoys sharing children's books with his mother and playing simple tablet games that involve tapping and dragging objects. During previous visits to the family doctor, he has cried and resisted participating in unfamiliar activities. Holding his favourite stuffed animal or playing a familiar game helps him settle. He feels proud when he receives a sticker at the end of the appointment, recognizing his participation.
- **Goals:**
  - Understand what will happen during the eye appointment, including what procedures might feel like, and feel less worried about unfamiliar equipment/procedures.
  - Know what he can do during each preparation activity and examination procedure, to make it less intimidating and go smoothly.
  - Feel proud of participating, and show his mother the activities he completed.
- **Frustrations:**
  - Instructions presented only as text, which he cannot read independently.
  - Loud environments that make him feel overwhelmed or make spoken instructions hard to hear.
  - Small or similar symbols that he struggles to distinguish between in games.
  - Unfamiliar controls that give him no clear guidance when he makes a mistake.
- **Technical Ability:** Noah is familiar with basic tapping and swiping on the family tablet. He recognizes some familiar symbols, but cannot read instructions independently. He follows short spoken instructions and visual demonstrations, with help from his mother when something is too unfamiliar. He needs guidance for complex gestures or navigation with many steps.

4.2 Persona 2: Mother of Paediatric Ophthalmology Patient

- **Name:** Sarah.
- **Description:** Sarah is a 33-year-old personal support worker (PSW) at an assisted living facility and the mother of four-year-old twins Noah and Nora. She often works 12–13-hour shifts and balances time with her children and errands on her full days off. Together, they read children's books, play games on the family tablet, watch their favourite shows, and visit nearby parks and recreation facilities. During these activities, Sarah noticed Noah's difficulty seeing and raised her concerns with the family doctor, leading to referral to an ophthalmologist. Her own childhood experience of struggling to see in the classroom prior to receiving glasses makes her particularly concerned about addressing her son's difficulties. She finds it stressful when Noah becomes distressed during doctor's office visits, and is unsure how to prepare him for unfamiliar procedures. She has observed that his favourite stuffed animal and tablet games help him settle. Before his first eye appointment, she wants to introduce preparation activities on the tablet, play alongside him, and help explain anything he does not understand.

- **Goals:**
  - Help Noah understand what to expect and feel more comfortable participating in his appointment.
  - Feel more confident supporting Noah when he becomes distressed.
  - Fit preparation into the available time around her work shifts and other responsibilities.
- **Frustrations:**
  - Products for her children with unclear setup instructions or unexplained terminology.
  - Being unsure how to explain unfamiliar things in language Noah understands.
  - Activities that require lengthy involvement or continuous adult guidance, making it difficult to fit into her schedule.
  - Account creation and forgotten passwords that delay getting started with applications for her children.
- **Technical Ability:** Sarah installs and uses mobile applications for banking, work scheduling, navigation, and casual games. She can set up activities on the family tablet, but becomes frustrated by complicated setup and managing multiple accounts. Although she works in a care setting, she has no knowledge of paediatric ophthalmology and needs clear explanations of unfamiliar eye examination procedures as well.

4.3 Persona 3: Paediatric Ophthalmologist

- **Name:** Daniel.
- **Description:** Daniel is a 45-year-old paediatric ophthalmologist working at a clinic associated with the local children's hospital. He regularly examines children aged three to seven, including patients who are anxious or unfamiliar with examination procedures. Although experienced in explaining examinations to children, he finds that a brief explanation immediately before a procedure can be insufficient to address their uncertainty. For example, a child encountering equipment for the first time may become distressed and need reassurance before participating. Daniel wants to give each child the support they need while keeping his appointments on schedule. During development, he reviews the preparation activities for clinical accuracy, including the stages shown and how sensations are described. He also may briefly demonstrate an activity during a consultation.
- **Goals:**
  - Help children arrive familiar with common procedures and the instructions they may be asked to follow.
  - Support children's participation during examinations and reduce delays.
- **Frustrations:**
  - Limited appointment time to explain procedures and provide each child with the reassurance they need.
  - Explanations that use language children find difficult to understand, requiring him to explain the same instructions in different ways.
  - Preparation materials that miss important steps or give children inaccurate expectations about what a procedure might feel like.
  - Difficulty finding and demonstrating a relevant preparation activity during consultation.

- **Technical Ability:** Daniel is comfortable using standard clinical software at work and mobile applications in daily life. He can install and navigate the preparation application on a mobile device, review its activities, and demonstrate them to patients and parents. He needs application navigation that allows him to find a relevant activity without much setup or technical assistance.

5 Use Cases

Each use case lists the stakeholder requirements it addresses and the functional requirements that refine it.

The following use cases describe how children and caregivers use EyeCanDo to explore ophthalmology preparation games through a journey of levels. All procedure game levels remain accessible regardless of whether earlier levels have been completed. Children may choose a character, earn journey points, receive in-game coins or cosmetic items, and use owned items to customize their character. The personal scoreboard shows the child's own progress, not a ranking against other children. The reward amounts, character designs, available accessories, and specific gameplay interactions described below are initial design proposals that will be refined with the project team and clinical supervisors.

5.1 UC1: Complete Initial Application Setup

Primary Actor

Caregiver

Preconditions

- The application is installed on a supported device.
- The application is being opened for the first time, and no caregiver acknowledgment has been recorded.

Main Success Scenario

1. The caregiver opens EyeCanDo.
2. The system displays a welcome screen explaining that the application helps children become familiar with eye examinations through interactive games.
3. The system displays a notice stating that the application is an educational preparation aid and not a substitute for professional medical advice, diagnosis, or treatment.
4. The caregiver reads the notice and selects the acknowledgment control.
5. The system records the acknowledgment locally.
6. The system displays an introduction to the journey, explaining that all procedure game levels are available and that children can earn points and cosmetic rewards as they play.
7. The caregiver continues to the home screen, where the child can choose a character and start exploring the journey.

Alternative Flows

- **Notice Not Acknowledged:** If the caregiver does not acknowledge the notice, the system does not complete initial setup.
- **Setup Interrupted:** If the application closes before acknowledgment is saved, the system displays the notice again on the next launch.
- **Returning User:** If acknowledgment was previously recorded, the system opens the home screen without repeating setup.

Postconditions

- The caregiver has been informed of the application's purpose and limitations.
- Acknowledgment is recorded locally, and the child can access the journey without creating an account.

Stakeholder Requirements

STR4, STR10, STR13, STR14.

Functional Requirements

FR1, FR2.

5.2 UC2: Choose a Journey Character

Primary Actor

Child

Preconditions

- Initial application setup has been completed.
- The application is displaying the home screen or character selection screen.

Main Success Scenario

1. The child selects the character button on the home screen.
2. The system displays the available starting characters with large visual previews.
3. The child selects a character from the gallery.
4. The system displays an enlarged preview of the selected character and a confirmation control.
5. The child confirms the selection.
6. The system saves the selected character identifier locally.
7. The system returns to the home screen and displays the selected character beside the journey entrance.

Alternative Flows

- **Try Another Character:** Before confirming, the child selects a different character; the system updates the preview.
- **Cancel Selection:** If the child returns home without confirming, the previously selected character remains active.
- **Change Later:** If a character was previously saved, the child can return to character selection and replace it without losing levels, points, coins, or owned accessories.

Postconditions

- The selected character appears in the child's journey.
- Game progress and reward balances remain unchanged.

Stakeholder Requirements

STR4, STR12, STR18.

Functional Requirements

FR31, FR32.

5.3 UC3: Select a Level on the Journey Map

Primary Actor

Child

Preconditions

- The application is open and initial setup is complete.
- The journey contains the Vision Testing, Eye-Drop Preparation, and Slit-Lamp Preparation levels.

Main Success Scenario

1. The child selects the journey button on the home screen.
2. The system displays a journey map containing the procedure levels in a suggested order, with a label and illustration for each.
3. The system shows each level's current status: Not Started, In Progress, or Completed.
4. The child selects the Slit-Lamp Preparation level, even though the Vision Testing and Eye-Drop levels are unfinished.
5. The system displays the slit-lamp level's title, an introductory illustration, and a play control.
6. The child selects play.
7. The system starts the slit-lamp level and presents its child-friendly instructions.

Alternative Flows

- **Select Another Level:** The child selects Vision Testing or Eye-Drop Preparation; the system opens that level's introduction.
- **Replay Completed Level:** The child selects a completed level; the system permits replay without resetting the completed status or reissuing the first-completion reward.
- **Return Home:** The child selects home before pressing play; no level status or reward balance changes.

Postconditions

- The selected preparation level is open.
- No earlier level had to be completed or unlocked first.

Stakeholder Requirements

STR1, STR3, STR4, STR7, STR8, STR9, STR11, STR16, STR18.

Functional Requirements

FR3, FR4, FR5.

5.4 UC4: View a Caregiver Procedure Guide

Primary Actor

Caregiver

Preconditions

- The application is open.
- The built-in caregiver guide contains explanations of the supported ophthalmology procedures.

Main Success Scenario

1. The caregiver selects the caregiver guide on the home screen.
2. The system lists Vision Testing, Eye-Drop Administration, and Slit-Lamp Examination.
3. The caregiver selects Eye-Drop Administration.
4. The system displays a clinician-reviewed explanation of why eye drops may be used and what a child may experience.
5. The system explains that the Eye-Drop Preparation level uses a virtual character to show a clinician administering drops.
6. The system displays the educational use disclaimer in the guide.
7. The caregiver returns to the guide menu or home screen.

Alternative Flows

- **Different Procedure:** The caregiver selects Vision Testing or Slit-Lamp Examination; the matching guide is displayed.
- **Exit Guide:** Returning home does not affect the child's game progress, score, or rewards.

Postconditions

- The caregiver has reviewed preparation information for the selected procedure.
- No diagnostic or treatment recommendation is produced.

Stakeholder Requirements

STR1, STR5, STR6, STR7, STR11, STR14.

Functional Requirements

FR1, FR39.

5.5 UC5: Play the Vision Testing Level

Primary Actor

Child

Preconditions

- The child has selected Vision Testing on the journey map.
- The level's introduction is displayed.

Main Success Scenario

1. The system plays a child-friendly introduction showing how a clinician may ask a child to identify symbols during a vision test.
2. The system explains the matching activity using narration and visual examples rather than requiring reading.
3. The system displays a reference symbol and several possible matching symbols.
4. The child selects the matching symbol or selects skip.
5. The system gives encouraging feedback on the selection or acknowledges the skipped round without penalty.

6. Steps 3–5 repeat with different symbols until every required matching round has been attempted or skipped.
7. The system displays the end-of-level screen and records the Vision Testing level as Completed.
8. The system initiates the first-completion reward process described in UC9.

Alternative Flows

- **Incorrect Match:** The system shows the matching symbol with narration and cues that do not rely on colour. The child selects it and receives encouraging feedback before the next round.
- **Skip Round:** The system advances to the next symbol without requiring a correct response.
- **Exit Early:** The child selects home before the final round. The system returns to the journey map and retains an In Progress status rather than marking the level Completed.

Postconditions

- If the end-of-level screen was reached, the level is marked Completed even when some rounds were skipped.
- The game does not measure visual acuity or provide clinical results.

Stakeholder Requirements

STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR15, STR17, STR18.

Functional Requirements

FR6, FR8, FR9, FR11, FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19.

5.6 UC6: Play the Eye-Drop Preparation Level

Primary Actor

Child

Preconditions

- The child has selected Eye-Drop Preparation on the journey map.
- The level's introduction is displayed.

Main Success Scenario

1. The system introduces a virtual patient character and explains that a clinician may place drops in a child's eyes.
2. The system illustrates a preparation action, such as having the virtual patient tip their head back.
3. The child uses the on-screen controls to help the virtual patient perform the demonstrated action, or selects skip.
4. The system responds with encouraging feedback or acknowledges the skipped action.
5. Steps 2–4 repeat for the remaining preparation actions.
6. The system presents an animation showing a clinical professional administering eye drops to the virtual patient.
7. After the sensation description, the child watches the animation or uses skip to bypass it if uncomfortable.

8. The system explains that the child may tell the clinician if they feel worried or uncomfortable during an examination.
9. The system marks the Eye-Drop Preparation level Completed and opens its end-of-level reward screen.
10. The system initiates the first-completion reward process described in UC9.

Alternative Flows

- **Incorrect Action:** The system shows a visual hint and allows a retry or skip.
- **Uncomfortable with Animation:** The child skips the administration animation and continues to the concluding message.
- **Exit Early:** The child selects home. The game stops and remains In Progress rather than Completed.

Postconditions

- If the child reaches the ending, the level is Completed regardless of skipped individual steps.
- The application does not instruct the child to administer real medication or carry out the procedure independently.

Stakeholder Requirements

STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR15, STR17, STR18.

Functional Requirements

FR6, FR8, FR9, FR10, FR11, FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19.

5.7 UC7: Play the Slit-Lamp Preparation Level

Primary Actor

Child

Preconditions

- The child has selected Slit-Lamp Preparation on the journey map.
- The level's introduction is displayed.

Main Success Scenario

1. The system introduces a simulated examination room and displays a slit-lamp device.
2. The system demonstrates how a virtual patient places their chin on the chin rest.
3. The child uses the on-screen controls to move the virtual patient's chin into place, or selects skip.
4. The system shows whether the virtual patient is positioned correctly or acknowledges the skipped exercise.
5. Steps 2–4 repeat for the forehead rest and other supported positioning exercises.
6. The system demonstrates that the patient may be asked to remain relatively still briefly while the clinician looks at their eyes.
7. The child completes a short virtual stillness exercise or chooses to skip it.
8. The system shows an encouraging examination ending animation and marks the level Completed.
9. The system initiates the first-completion reward process described in UC9.

Alternative Flows

- **Position Incorrect:** The system provides a visual prompt to reposition the virtual patient or skip the exercise.
- **Unable to Remain Still:** The child skips the virtual stillness exercise and proceeds without a penalty.
- **Exit Early:** The child selects home; the system returns to the journey map and does not mark the level Completed.

Postconditions

- If the ending is reached, the level is marked Completed.
- The game does not measure the child's head movement or produce clinical measurements. Camera tracking is not assumed.

Stakeholder Requirements

STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR15, STR17, STR18.

Functional Requirements

FR6, FR8, FR9, FR11, FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19.

5.8 UC8: Exit a Level Before Completion

Primary Actor

Child

Preconditions

- The child is playing a procedure game level and has not reached its ending.

Main Success Scenario

1. During a Vision Testing matching round, the child decides to stop playing.
2. The child selects the visible home control.
3. The system stops the matching interaction, its narration, animations, and game-specific input processing.
4. The system records the Vision Testing level as In Progress unless it had previously been completed.
5. The system returns the child to the journey map.
6. The system displays the Vision Testing level with its current status while leaving every other level accessible.
7. The child's existing journey points, coin balance, and owned character accessories remain unchanged.

Alternative Flows

- **Other Level:** The same exit behaviour applies while playing the Eye-Drop or Slit-Lamp Preparation level.
- **Animation Playing:** If the child selects home during an animation, the system stops the animation and audio before displaying the journey map.
- **Replay:** When the child reselects an interrupted level, it restarts at its introduction. The level is not falsely recorded as Completed.
- **Device Interruption:** If the application is closed during a level, reopening it restores access to the journey and preserves previously saved progress and rewards.

Postconditions

- The interrupted level is no longer displayed or processing interactions.
- The child's saved rewards and other completed levels remain unchanged.
- No first-completion reward is issued for the interruption.

Stakeholder Requirements

STR2, STR7, STR9.

Functional Requirements

FR20, FR21, FR22, FR23, FR24, FR25, FR26.

5.9 UC9: Receive a Level Completion Reward

Primary Actor

Child

Preconditions

- The child has reached the ending of a procedure game level after attempting or skipping its exercises.
- The system has a predefined first-completion reward for the level.

Main Success Scenario

1. The child reaches the end of the Vision Testing level.
2. The system records the level as Completed and checks whether its first-completion reward was previously claimed.
3. Because the reward is unclaimed, the system adds the level's journey points to the child's personal total.
4. The system awards the Vision Testing level's predefined quantity of in-game coins and adds them to the child's coin balance.
5. The system displays a reward screen showing the journey points earned, the coins received, and the updated coin balance.
6. The child selects continue.
7. The system returns to the journey map and displays the completed level with its reward recorded.

Alternative Flows

- **Direct Cosmetic Reward:** If the level's reward is a cosmetic item rather than coins (for example, a star-shaped accessory for the Eye-Drop level), the system adds that item directly to the child's inventory, displays its preview, and offers a way to equip it. A level's reward may include coins, a cosmetic item, both, or neither.
- **Skipped Exercises:** If the child reached the ending after skipping some exercises, the system still awards the predefined participation-based reward.
- **Replay Completed Level:** If the child completes an already rewarded level again, the system congratulates the child but does not award duplicate points, coins, or cosmetic items.
- **Exit Before Ending:** If the child exits before the ending, the system does not issue the first-completion reward.

Postconditions

- The first-completion reward is recorded only once for the level.
- Journey points and coin balance or owned cosmetic items are updated locally as appropriate.
- Rewards represent participation, not visual ability or clinical readiness.

Stakeholder Requirements

STR3, STR8, STR9, STR16.

Functional Requirements

FR5, FR28, FR29.

5.10 UC10: View the Personal Journey Scoreboard

Primary Actor

Child

Preconditions

- The application is open and journey data is available locally.

Main Success Scenario

1. The child selects the scoreboard button from the journey map.
2. The system displays the selected character and the child's optional nickname, if one has been entered.
3. The system reads the saved progress for the Vision Testing, Eye-Drop Preparation, and Slit-Lamp Preparation levels.
4. For each level, the system displays its name, level number, status (Not Started, In Progress, or Completed), and points earned.
5. Step 4 repeats until all journey levels are displayed.
6. The system calculates and displays total journey points and the number of completed levels out of the total available.
7. The system separately displays the child's current spendable coin balance.
8. The child reviews the scoreboard and selects home to return to the journey map.

Alternative Flows

- **No Levels Played:** The system displays every level as Not Started, zero completed levels, and zero earned points.
- **Level Interrupted:** An interrupted level appears as In Progress unless it was already completed earlier.
- **Coins Spent:** If the child purchased an accessory, the coin balance is lower, but previously earned journey points and completed level counts remain unchanged.
- **All Levels Completed:** The system shows the full journey as Completed and displays a celebratory final scoreboard.

Postconditions

- The child has viewed a personal scoreboard without comparing results to other users.

- Viewing the scoreboard does not change progress or balances.
- Points represent participation, not clinical performance.

Stakeholder Requirements

STR3, STR8, STR9, STR16.

Functional Requirements

FR30.

5.11 UC11: Purchase a Character Accessory with Coins

Primary Actor

Child

Preconditions

- A character is selected and the application is open.
- The child has earned enough in-game coins for at least one accessory in the cosmetic shop.

Main Success Scenario

1. The child selects the character shop from the journey map.
2. The system displays cosmetic accessories, such as hats and glasses, with visual previews and prices in in-game coins.
3. The child selects an unowned hat.
4. The system displays the hat on the selected character, its coin price, and the child's current coin balance.
5. The child selects the purchase control.
6. The system verifies that the child has enough coins and that the selected hat has not already been purchased.
7. The system subtracts the hat's coin price from the spendable coin balance and adds the hat to the child's inventory.
8. The system displays a purchase confirmation and offers an option to equip the new hat.

Alternative Flows

- **Insufficient Coins:** If the coin balance is too low, the system displays the shortage and does not deduct coins or add the item to the inventory.
- **Item Already Owned:** If the child selects an owned item, the system identifies it as owned and offers to equip it rather than charging coins again.
- **Cancel Purchase:** If the child returns to the shop without confirming, no coins or inventory items change.
- **Direct Reward Item:** An accessory received directly from a completed level appears as owned and is not purchased again.

Postconditions

- The purchased accessory is available in the child's inventory.

- The coin balance decreases by the stated price, but total journey points and completed levels do not change.
- No real money payment, account, or Internet connection is used.

Stakeholder Requirements

STR3, STR8, STR12, STR15, STR16.

Functional Requirements

FR32, FR33, FR34, FR35, FR37.

5.12 UC12: Customize a Character with an Owned Accessory

Primary Actor

Child

Preconditions

- A character is selected.
- At least one cosmetic accessory has been purchased with coins or received directly as a level reward.

Main Success Scenario

1. The child selects the character customization button.
2. The system displays the current character and its equipped accessories.
3. The child opens the owned accessories collection.
4. The system shows a hat previously earned as a direct level reward alongside other owned items.
5. The child selects the hat.
6. The system previews the hat on the character.
7. The child selects equip.
8. The system updates the character's appearance and saves the equipped hat locally.
9. The child returns to the journey, where the character appears wearing the chosen hat.

Alternative Flows

- **Unequip:** The child selects remove for an equipped accessory; the item remains owned and can be equipped later.
- **Replace Accessory:** If another item occupies the same accessory category, the system replaces the displayed item without deleting either item from the inventory.
- **Locked Accessory:** If the child selects an item not owned, the system does not equip it and shows how it can be obtained through the shop or journey rewards.
- **Cancel Preview:** Returning before selecting equip preserves the character's existing appearance.

Postconditions

- The selected owned accessory is equipped and saved locally.
- Points, coin balance, and game completion statuses are unchanged by equipping or removing an item.

Stakeholder Requirements

STR3, STR8, STR12.

Functional Requirements

FR32, FR36.

5.13 UC13: Change a Saved Nickname

Primary Actor

Caregiver

Preconditions

- The application is open.
- The caregiver can access application settings.
- An optional nickname has previously been saved.

Main Success Scenario

1. The caregiver selects settings on the home screen.
2. The system displays the saved nickname and the available audio, display, and accessibility preferences.
3. The caregiver selects the nickname field.
4. The system allows the existing text to be edited.
5. The caregiver enters a new nickname of their choice and selects save.
6. The system stores the new nickname locally and displays it in the settings field.
7. The caregiver returns to the home screen.
8. The system displays the new nickname wherever personalization is used, including the personal scoreboard.

Alternative Flows

- **Clear Nickname:** The caregiver removes all text; the system uses default, non-personalized wording.
- **Cancel Edit:** If the caregiver cancels without saving, the previously saved nickname remains unchanged.
- **Other Preference:** The caregiver may change audio, display, or accessibility settings instead.

Postconditions

- The nickname is updated locally and persists across restarts.
- The selected character, purchased accessories, points, coin balance, and game completion statuses are unchanged.
- The nickname is not an account identifier or login credential.

Stakeholder Requirements

STR3, STR13, STR16, STR17.

Functional Requirements

FR7, FR14, FR38, FR40, FR41.

5.14 UC14: Reset the Local Journey Data

Primary Actor

Caregiver

Preconditions

- The application is open.
- The caregiver can access data management settings.

Main Success Scenario

1. The caregiver opens settings and selects reset journey data.
2. The system explains that resetting will delete the saved nickname, selected character, equipped and owned accessories, coin balance, journey points, and game progress.
3. The system asks the caregiver to confirm the reset.
4. The caregiver confirms.
5. The system deletes the listed locally stored information.
6. The system restores all level statuses to Not Started, total points and coin balance to zero, and the character to its initial unselected or default state.
7. The system returns to the home screen with all procedure game levels still available.

Alternative Flows

- **Cancel Reset:** If the caregiver cancels, no saved information changes.
- **No Progress:** If the journey has not been played, the system still allows the caregiver to cancel or confirm.
- **Acknowledgment Reset:** If the caregiver selects the broader option to clear all local application data, the stored educational notice acknowledgment is also erased. The system shows the medical notice again before the home screen.

Postconditions

- The selected locally stored journey data has been erased.
- Built-in game content and character catalogues remain installed and accessible; no network download is required.

Stakeholder Requirements

STR13, STR16.

Functional Requirements

FR38, FR42, FR43.

5.15 UC15: Play and Earn Journey Rewards Offline

Primary Actor

Child

Preconditions

- The application was installed successfully and its initial setup was previously completed.

- The application is closed and the device is in airplane mode, with no Internet connection.

Main Success Scenario

1. The child opens EyeCanDo without Internet access.
2. The system loads locally stored character selection, journey progress, points, coin balance, and owned accessories.
3. The system displays the home screen and journey map, with all three preparation levels available.
4. The child selects the Vision Testing level.
5. The system loads its introduction, narration, and symbol matching rounds from locally available content.
6. The child matches or skips the displayed symbols; the system gives feedback and proceeds through the rounds.
7. The child reaches the ending of the level.
8. The system updates the level status and awards any unclaimed first-completion journey points and coins locally.
9. The child opens the personal scoreboard and sees the updated level status, points, and coin balance.
10. The child opens character customization and sees previously acquired accessories without network access.

Alternative Flows

- **Other Level:** The child selects Eye-Drop or Slit-Lamp Preparation; its built-in content also works offline.
- **Offline Purchase:** The child purchases a cosmetic item using sufficient locally recorded coins; the balance and inventory update without contacting an external server.
- **Exit Early:** The child exits before the level ending; the level remains In Progress and no first-completion reward is issued.
- **Connection Lost Midgame:** If a connection is disabled while the application is already running, gameplay, scoring, and locally stored rewards continue functioning.

Postconditions

- Journey progress and any earned rewards are saved locally.
- Levels, the personal scoreboard, and character customization function without an Internet connection or external account.

Stakeholder Requirements

STR4, STR10, STR13, STR15.

Functional Requirements

FR26, FR27.

6 Functional Requirements

The functional requirements state what EyeCanDo shall do. They refine the use cases of Section 5.

EyeCanDo presents a single journey. The journey has one level for each procedure, such as vision testing, eye drops, or the slit-lamp examination. Any level can be played at any time. Finishing a level for the first time gives a level reward.

This section has two parts. Section 6.1 first defines a formal model of the player's progress and rewards. Section 6.2 then states the functional requirements. A requirement may use a name from the model, and it then cites the definition or theorem it relies on. Where a requirement and the model describe the same behaviour, the model is authoritative.

The format of each level is not yet decided. The model and the requirements therefore describe the journey and the progress through its levels, and not what happens inside a level.

Several requirements come from the safety and security requirements of the Hazard Analysis. Those requirements are stated here as functional requirements when they describe what the system does, and in Section 8 when they describe a quality of the system or the release process. Each requirement that comes from the Hazard Analysis has a *Hazard Analysis* line. The line names the safety and security requirement (SR) and the hazard (H) it addresses, using the numbering of the Hazard Analysis. Each SR and H number links to that item in the published Hazard Analysis. Table 9 traces each safety and security requirement to the functional requirements.

6.1 Formal Model

This section formalizes the most critical part of EyeCanDo. This is the child's progress through the levels and the rewards for finishing them. It is used every time a child plays, and an error here harms an anxious child. Examples are a locked level, a reward that is lost or given twice, or progress that disappears. The shop, the characters and the accessories are not modelled. Their requirements are simple enough to state in words (Section 6.2.8).

Definitions (D1, D2, ...) fix the model. Theorems (T1, T2, ...) state the properties that the requirements depend on. Each theorem has a short proof and lists the requirements it specifies.

Notation. Names in italics, such as *coins*, are names in the model. Values in a sans-serif font, such as Completed, are values. $\mathbb{N}$ is the natural numbers, including 0. A primed name, such as $coins'$, is the value after an operation.

Game content. The team defines the game content before release, and it does not change while the application runs. *Levels* is the finite, non-empty set of levels, one for each procedure (Procedure Game in the data dictionary). For each level, $points(level) \in \mathbb{N}$ and $coinReward(level) \in \mathbb{N}$ are the journey points and coins for finishing it the first time (Reward Definition). A level reward may also include an accessory. It is given at the same moment as the coins, so every result below about coins also holds for it.

Definition D1: Level status

$$Status = \{\mathsf{NotStarted}, \mathsf{InProgress}, \mathsf{Completed}\} \tag{E1}$$

NotStarted means the level has never been started. InProgress means it has been started but not finished. Completed means it has been finished at least once.

Definition D2: Player state

The player state is the pair $(status, coins)$, where $status : Levels \to Status$ and $coins \in \mathbb{N}$.

In the *new player state*, $status(level) = \mathsf{NotStarted}$ for every level, and $coins = 0$.

These are completion_status and coin_balance in the data dictionary. A level's reward is given exactly when it becomes Completed, so reward_claimed is true exactly when the status is Completed.

Definition D3: Scoreboard values

$$totalPoints = \sum_{\substack{level \in Levels \\ status(level) = \mathsf{Completed}}} points(level) \tag{E2}$$

$$completedLevels = |\{level \in Levels \mid status(level) = \mathsf{Completed}\}| \tag{E3}$$

These values are calculated from the player state and are not stored.

Definition D4: Operations

Operation	Guard	Effect
$start(level)$	always	if $status(level) \neq \mathsf{Completed}$, then $status'(level) = \mathsf{InProgress}$
$finish(level)$	$status(level) = \mathsf{InProgress}$	$status'(level) = \mathsf{Completed}$ and $coins' = coins + coinReward(level)$
$exit(level)$	always	none
$reset$	always	the new player state

start is used when the child starts a level, *finish* when the child reaches the end of a level, *exit* when the child leaves a level early, and *reset* when the caregiver resets the journey. If the guard is false, nothing changes. A part of the player state that the effect does not name keeps its value.

Theorem T1: Every level can always be started

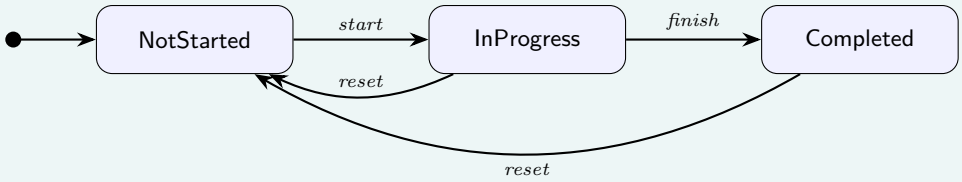
For every level and every player state, $start(level)$ takes effect. Afterwards the level is Completed if it was Completed before, and InProgress otherwise.

Proof. The guard of *start* is always true, and its effect is as stated (D4). □

Specifies: FR4, FR5.

Theorem T2: Level status state machine

The status of a level changes only along the arrows below. In particular, a Completed level stops being Completed only by *reset*.



Proof. Read the effects in D4. *start* changes only a status that is not Completed, to InProgress. *finish* changes only InProgress, to Completed. *reset* makes every level NotStarted. *exit* changes nothing. □

Specifies: FR4, FR5, FR11, FR22, FR42.

Theorem T3: Each reward is given once

Between two resets, the first *finish*(*level*) after the level is started adds *points*(*level*) to *totalPoints* and *coinReward*(*level*) to *coins*. Every later *finish*(*level*) changes nothing.

Proof. The first *finish* finds the level InProgress, so it makes the level Completed and adds the coins (D4). By equation (E2), *totalPoints* now counts the level. By T2, the level stays Completed until a reset, so the guard of every later *finish* is false. □

Specifies: FR11, FR18, FR28, FR29.

Theorem T4: Progress is lost only by a reset

For *start*, *finish* and *exit*, a Completed level stays Completed, $totalPoints' \geq totalPoints$, and $coins' \geq coins$. In particular, *exit* changes nothing.

Proof. The first part is T2. Then *totalPoints* cannot decrease, by equation (E2). Only *finish* changes *coins*, and it only adds to them. *exit* has no effect (D4). □

Spending coins in the shop is not one of these operations, so this theorem does not cover it.

Specifies: FR5, FR18, FR20, FR22, FR24.

6.2 Requirements

This subsection states the functional requirements, grouped by feature. Each requirement has a rationale and a *Traces* line. The *Traces* line lists the use cases the requirement refines and the definitions and theorems of Section 6.1 it relies on. A requirement that helps satisfy non-functional requirements also has a *Related NFRs* line that lists them. A requirement that comes from the Hazard Analysis also has a *Hazard Analysis* line.

6.2.1 Setup

FR1: Before first use, the system shall display a medical notice to the caregiver. The notice shall state that EyeCanDo is an educational preparation aid and does not replace professional medical advice, diagnosis, or treatment. The system shall not open the home screen until the caregiver acknowledges the notice. The notice shall also be available from the caregiver procedure guide (FR39).

Rationale: A caregiver could mistake the procedure content for medical guidance. Requiring an acknowledgement makes sure the caregiver has seen what the application is for before the child uses it.

Traces: UC1, UC4.

Related NFRs: NFR32.

Hazard Analysis: SR-10 (H2-8).

FR2: When the caregiver acknowledges the medical notice (FR1), the system shall store the acknowledgement locally. On later launches with a stored acknowledgement, the system shall open the home screen without displaying the notice.

Rationale: The caregiver only needs to read the notice once. Showing it on every launch would slow down every session and teach caregivers to dismiss it without reading.

Traces: UC1.

Hazard Analysis: SR-10 (H2-8).

6.2.2 Journey Map

FR3: The system shall display a journey map that shows every level, one for each procedure, in order of level number. Each level shall show its name, an illustration, and its status (D1: NotStarted, InProgress, or Completed).

Rationale: The journey map is how the child chooses what to play. The status of each level shows the child and caregiver what has been done so far.

Traces: UC3, D1.

Related NFRs: NFR5, NFR6, NFR10.

FR4: The system shall allow the child to select any level at any time. The status of the other levels, the journey points, and the rewards shall not restrict this choice (T1). When the child selects a level, the system shall start it from the beginning and apply *start* (D4). If the level was not Completed, its status shall become InProgress.

Rationale: The procedures at an appointment can change at short notice. A locked level could stop a child from preparing for the procedure they actually need. Starting from the beginning gives the child the full context of the procedure each time. Recording InProgress shows on the journey map that the level has been tried.

Traces: UC3, D4, T1, T2.

Related NFRs: NFR6.

Hazard Analysis: SR-11 (H2-11).

FR5: The system shall allow the child to replay a Completed level. The level shall remain Completed

during and after the replay.

Rationale: Repeating a level helps a child become familiar with a procedure. A replay must not undo the progress the child has already made.

Traces: UC3, UC9, T1, T2, T4.

6.2.3 Level Content

FR6: Each level shall prepare the child for one ophthalmology procedure. It shall present at least one part of that procedure, such as a step of the procedure, a piece of equipment, or an instruction the child may be given.

Rationale: The purpose of the journey is preparation. A level that does not relate to a procedure adds play time without preparing the child. The format of each level is not yet decided, so this requirement states only what a level must convey.

Traces: UC5, UC6, UC7.

Related NFRs: NFR22.

FR7: The age group, which the caregiver sets (FR41), shall be either 3 to 4 years or 5 to 7 years, or not set. When an age group is set and a level provides content for that age group, the system shall present that content. Otherwise, the system shall present the level's default content.

Rationale: A three-year-old and a seven-year-old understand explanations differently. Content for the child's age group is easier to follow. The supervisor discussed whether content should vary by age (ER2), and the two groups match the split used in NFR5. Default content means that every level always has content to show.

Traces: UC13.

Hazard Analysis: SR-24 (H2-3).

FR8: Before a level shows a part of a procedure that the child may feel, such as cold, stinging, a bright light, pressure, or blurry vision, the system shall describe that sensation to the child. The description shall not be skippable, even when the child skips the exercise or animation that shows that part (FR18). The system shall not state or imply that a procedure will not hurt or will not be uncomfortable.

Rationale: A child who has been told about a sensation is not surprised by it at the appointment. A promise that nothing will be felt would be broken at the clinic, and the child would learn to distrust the people who prepared them.

Traces: UC5, UC6, UC7.

Related NFRs: NFR2, NFR22.

Hazard Analysis: SR-8 (H2-6).

FR9: Each level shall tell the child, in narration and pictures, that their doctor may do things a little differently from the game.

Rationale: Procedures vary between patients, clinicians, and clinics. A child who expects small differences is less likely to be upset by them at the appointment.

Traces: UC5, UC6, UC7.

Related NFRs: NFR22.

Hazard Analysis: SR-6 (H2-5).

FR10: Before the end of each level, the system shall tell the child that they can tell the doctor if they feel worried or uncomfortable.

Rationale: Knowing that they can speak up gives the child some control during the examination, which can reduce anxiety.

Traces: UC6.

FR11: When the child reaches the end of a level, the system shall display an end-of-level screen and apply *finish* (D4). The level's status shall become Completed.

Rationale: A clear ending gives the child a sense of achievement. Recording the level as Completed lets the journey map and scoreboard show it.

Traces: UC5, UC6, UC7, D4, T3, T2.

6.2.4 Instructions and Feedback

FR12: The system shall give every instruction to the child as narration together with a visual demonstration. No level shall require the child to read.

Rationale: Most children aged 3 to 7 cannot read independently. Narration with a demonstration lets them follow without an adult reading to them.

Traces: UC5, UC6, UC7.

Related NFRs: NFR5, NFR8, NFR9, NFR14.

Hazard Analysis: SR-24 (H2-3).

FR13: When an exercise asks the child to perform an action, such as moving a virtual patient into place, and the child does something else, the system shall give an encouraging hint and let the child try again. A wrong answer to a question is handled by FR16 instead.

Rationale: A hint lets the child continue without help from an adult. Letting the child try again keeps the focus on learning rather than on getting it right the first time.

Traces: UC5, UC6, UC7.

Related NFRs: NFR5.

Hazard Analysis: SR-12 (H2-12).

FR14: The visual cues that accompany every instruction (FR12) shall be shown in every state of a level. The child shall not be able to turn them off. Any setting that changes them shall be available only in the caregiver settings (FR38).

Rationale: Visual cues are the only guidance for a child who cannot hear the narration. A child could turn them off by accident and then be unable to continue.

Traces: UC5, UC6, UC7, UC13.

Related NFRs: NFR9.

Hazard Analysis: SR-19 (H2-15).

FR15: When narration cannot be heard because the device is muted or its volume is off, the system shall continue to give every instruction through pictures, icons, and demonstrations. The system shall also show a prompt for the caregiver that explains how to turn the sound on. The prompt shall not stop the child from playing.

Rationale: Turning the sound down must not cut the child off from the preparation content. Deaf children and children in a quiet waiting room also need to be able to play without sound.

Traces: UC5, UC6, UC7.

Related NFRs: NFR9, NFR15.

Hazard Analysis: SR-14 (H2-16), SR-18 (H2-14).

FR16: When the child selects an incorrect answer to a question, the system shall show every correct answer and mark it with narration and a visual cue. The child shall then be able to select a correct answer and receive encouraging feedback before the level continues.

Rationale: A child who leaves a question unsure of the answer may stay worried about it. Letting the child select the correct answer ends the question on a success. How quickly the answer appears and how its cues work without colour are qualities of this behaviour, given in NFR12 and NFR10.

Traces: UC5, UC6, UC7.

Related NFRs: NFR9, NFR10, NFR12.

Hazard Analysis: SR-16 (H2-10).

FR17: Every question shown to the child shall have at least one correct answer.

Rationale: A question that cannot be answered correctly would leave the child stuck and confused, and FR16 could not be met. How this is checked before release, for example by a check when the application is built, is a verification choice for the VnV Plan.

Traces: UC5, UC6, UC7.

Hazard Analysis: SR-17 (H2-10).

FR18: The system shall allow the child to skip any exercise or animation in a level, except a sensation description (FR8). Skipping an exercise or giving an incorrect response shall not produce negative feedback and shall not remove any points, coins, or accessories. It shall not prevent the child from reaching the end of the level.

Rationale: Penalties would make the game a new source of stress. Letting the child skip an exercise lets them explore a procedure at their own pace. The reward for finishing a level does not depend on the answers (T3).

Traces: UC5, UC6, UC7, T3, T4.

Hazard Analysis: SR-12 (H2-12).

FR19: During a screen transition, the system shall act on at most one navigation input. Further touches before the next screen is shown shall be ignored. While the narration of procedure information is playing, touches outside the skip control, the home control, and the restart control shall not skip or shorten it. The home and restart controls shall keep working during narration (FR20, FR24). The skip control shall act only after it has been held for `SKIP_HOLD_MIN_S` second, and it shall not skip

a sensation description (FR8).

Rationale: Young children often tap the screen many times. Without this rule, extra taps could skip the explanation of a procedure that the child needs to hear. A hold separates a deliberate skip from a repeated tap, which a rule about "accidental" touches alone could not. The home and restart controls are exempt so that a child who is upset can still leave the level with one touch (NFR4).

Traces: UC5, UC6, UC7.

Related NFRs: NFR12.

Hazard Analysis: SR-3 (H2-1).

6.2.5 Exiting and Interruptions

FR20: When the child selects the home control during a level, the system shall end the level, apply *exit* (D4), and return to the journey map.

Rationale: A child who is upset or bored must be able to leave at any time. Returning to the journey map lets them choose another level.

Traces: UC8, D4, T4.

Related NFRs: NFR4.

Hazard Analysis: SR-4 (H2-2).

FR21: When a level ends early, the system shall stop all audio, animation, and visual effects of that level before the journey map is shown. None of them shall continue on the journey map.

Rationale: Sounds or animations from a level the child has left could confuse or upset them. Stopping everything makes leaving a level feel complete.

Traces: UC8, D4.

Related NFRs: NFR4.

Hazard Analysis: SR-13 (H2-13).

FR22: Leaving a level before its end shall not change the player state (D2). In particular, it shall not give a reward or change the status of any level.

Rationale: Leaving early is not a failure. If leaving cost the child progress, a child who is upset might feel pressure to keep playing.

Traces: UC8, D2, T2, T4.

Hazard Analysis: SR-4 (H2-2).

FR23: When the child starts a level that was left before its end, or that was interrupted because the application closed, the level shall start again from the beginning.

Rationale: Restarting from the beginning is simpler for a young child than resuming in the middle of a level they may not remember.

Traces: UC8.

Hazard Analysis: SR-4 (H2-2).

FR24: From every screen of a level, the system shall offer a way to restart the level and a way to return to the journey map. Input that the level does not expect shall be ignored and shall not stop the level from continuing. Restarting a level shall keep all saved progress and rewards.

Rationale: An unexpected input or an unfinished action must never leave the child in a state they cannot get out of. A restart that kept nothing would make the child lose what they had already earned.

Traces: UC8, T4.

Related NFRs: NFR4, NFR17.

Hazard Analysis: SR-4 (H2-2).

FR25: When the device sends the application to the background, or when an interruption such as a phone call or a notification takes over the audio or the screen, the system shall pause all audio and animation. When the interruption ends, the system shall continue from where it paused.

Rationale: A phone call or a locked screen should not make the child miss part of a level, and sound should not keep playing in the background.

Traces: UC8.

Related NFRs: NFR17.

Hazard Analysis: SR-18 (H2-14), SR-25 (H2-17).

6.2.6 Saving and Restoring Progress

FR26: The system shall save the player state (D2), the owned and equipped accessories, and the selected character on the device each time one of them changes, before processing the next input. A save that fails or is interrupted shall leave the previously saved player state unchanged. When the application is launched again, the system shall restore the saved player state.

Rationale: The child's progress and rewards must survive the application closing. Saving after every change means at most the current input can be lost. Keeping the previous save when a save fails means a failure never replaces valid progress with damaged data.

Traces: UC8, UC15, D2.

Related NFRs: NFR17, NFR19, NFR26.

Hazard Analysis: SR-20 (H2-18).

FR27: When the saved data is missing, cannot be read, or is not valid, the system shall replace it with the data of a new installation and continue without failing. That data is the new player state (D2), no accessories, and no selected character.

Rationale: A crash on launch would leave the child with no access to any level. Starting again from a new player state keeps every level available.

Traces: UC15, D2.

Related NFRs: NFR17.

Hazard Analysis: SR-20 (H2-18).

6.2.7 Rewards and Scoreboard

FR28: When a level is finished for the first time, the system shall give its level reward. The reward is the level's journey points, together with its coins and its accessory if the level's Reward Definition includes them. The system shall then display a reward screen showing what was earned.

Rationale: Rewards are for reaching the end of a level, not for correct answers. A child who needed hints receives the same reward, so no child is left behind. Defining the reward per level, as coins, an accessory, both, or neither, lets the team tune the rewards without changing the requirement.

Traces: UC9, D4, T3.

Related NFRs: NFR7, NFR19.

Hazard Analysis: SR-12 (H2-12).

FR29: When a level whose reward has already been given is finished again, the system shall congratulate the child without giving points, coins, or accessories.

Rationale: Giving the same reward again would encourage the child to replay one level instead of exploring the journey, and would make the totals on the scoreboard meaningless.

Traces: UC9, T3.

Related NFRs: NFR7.

FR30: The system shall display a personal scoreboard. For each level, it shall show the name, the status, and the points earned. It shall also show the values of D3: the total journey points, the number of completed levels out of the total, and the coin balance. It shall show the selected character and the nickname if they are set. It shall not compare the child with other users.

Rationale: The scoreboard shows the child what they have achieved. Comparing children would add pressure, which works against the goal of reducing anxiety.

Traces: UC10, D3.

6.2.8 Characters and Accessories

FR31: The system shall display every character the child can choose. When the child confirms a character, the system shall select it. Choosing a character shall change only the selected character. When no character is selected, the system shall use a default character.

Rationale: Choosing a character gives the child a sense of ownership of the journey. Changing characters must not cost the child any progress or rewards. A default character means that a child who has not chosen one, or whose data was reset (FR27, FR42), is never blocked from a level.

Traces: UC2.

FR32: The characters shall vary in skin tone, hair texture and style, body type, and gender expression. They shall include options with accessibility and medical devices, such as glasses, an eye patch, or a wheelchair. The accessories shall include items from a range of cultures and religions.

Rationale: A child who can see themselves in a character feels that the game, and the clinic it prepares them for, is a place where they belong.

Traces: UC2, UC11, UC12.

Related NFRs: NFR31.

Hazard Analysis: SR-15 (H2-9).

FR33: The system shall display a shop of every accessory that has a price. Each accessory shall show a preview on the selected character, its price, and whether the child owns it.

Rationale: Seeing a preview and the price lets the child decide what to save coins for. Marking owned accessories prevents the child from trying to buy something twice.

Traces: UC11.

FR34: When the child confirms the purchase of an accessory that is not owned and costs at most the coin balance, the system shall subtract the price from the coin balance and add the accessory to the owned accessories.

Rationale: Buying turns the coins the child earned into a visible reward, which gives the child a reason to continue the journey.

Traces: UC11.

Related NFRs: NFR19.

FR35: When the child tries to buy an accessory that is already owned or costs more than the coin balance, the system shall not change the player state. If the accessory is owned, the system shall offer to equip it. If it costs too much, the system shall show how many more coins are needed.

Rationale: Telling the child why the purchase did not happen avoids confusion. Not changing anything means the coin balance can never become negative.

Traces: UC11.

FR36: The system shall allow the child to equip any owned accessory and to unequip any equipped accessory. At most one accessory of each category shall be equipped at a time. Equipping or unequipping shall not remove any accessory from the owned accessories.

Rationale: One accessory per category keeps the character's appearance sensible, for example one hat at a time. Keeping owned accessories means the child never loses something they earned.

Traces: UC12.

FR37: The system shall not offer any purchase with real money.

Rationale: Coins are earned only by playing. Real-money purchases would not be appropriate for young children and would conflict with app store policies for children's applications.

Traces: UC11.

Related NFRs: NFR34.

6.2.9 Caregiver Features

FR38: The system shall place the settings behind a parental gate. The gate shall require the user to read a written instruction and complete it, such as answering an arithmetic question that is shown only in words. The gate shall not be passable by tapping pictures or by following narration. No setting shall be offered outside the settings.

Rationale: Settings such as the audio and the visual cues decide whether the child can follow a

level. A child who changed them by accident could be left unable to continue. Most children aged 3 to 7 cannot read (Section 2.4), so a gate that requires reading is one that an adult can complete and a child is unlikely to. App store policies for children's applications also expect a parental gate in front of adult functions. The settings are part of the caregiver area.

Traces: UC13, UC14.

Related NFRs: NFR34.

Hazard Analysis: SR-19 (H2-15).

FR39: The system shall provide a caregiver procedure guide. For each procedure, the guide shall explain what the child may experience and how the matching level presents it.

Rationale: Many caregivers do not know what happens during these procedures. The guide helps them answer the child's questions and support the child at the appointment.

Traces: UC4.

Related NFRs: NFR8, NFR22, NFR32.

FR40: The settings shall allow the caregiver to enter, change, or clear an optional nickname. When no nickname is set, the system shall use wording that is not personalized.

Rationale: A nickname makes the journey feel like the child's own. It is optional, and it does not need to be the child's real name, so the application does not need to collect identifying information.

Traces: UC13.

Related NFRs: NFR26, NFR29.

FR41: The settings shall allow the caregiver to change the age group (FR7), the narration volume, the music and sound-effect volume, the size of the visual cues, and whether motion effects are reduced. Each change shall take effect while the settings are still open, without restarting the application.

Rationale: Caregivers know what their child needs, such as a lower volume in a waiting room. Applying changes immediately lets the caregiver see the result before handing the device back.

Traces: UC13.

Related NFRs: NFR9.

Hazard Analysis: SR-14 (H2-16), SR-19 (H2-15), SR-24 (H2-3).

FR42: The settings shall allow the caregiver to reset the journey data. The system shall first state what will be erased and ask for confirmation. On confirmation, the system shall apply *reset* (D4), which returns the player state to the new player state (D2). It shall also remove all accessories, deselect the character, and clear the nickname. The notice acknowledgement shall be kept. The system shall then show the home screen. On cancellation, no data shall change.

Rationale: A reset lets a family start over, or pass the device to another child. The confirmation prevents accidental loss of progress. The acknowledgement is kept because the same caregiver has already read the notice.

Traces: UC14, D2, D4, T2.

Related NFRs: NFR29.

FR43: The settings shall allow the caregiver to clear all local application data after confirmation. This shall erase the data of FR42 and the notice acknowledgement. The system shall then show the medical notice (FR1) before the home screen.

Rationale: A caregiver may want to remove everything the application has stored, for example before giving the device to someone else.

Traces: UC14.

Related NFRs: NFR29.

7 Data Requirements

This section describes the information managed by EyeCanDo to support journey-based progression, character customization, game completion tracking, rewards, and a personal scoreboard. The application does not require patient accounts or integration with clinical information systems. All user data is maintained locally on the user's device.

Each ophthalmology procedure is represented by one preparation game, which acts as a level in the child's journey. Games may contain multiple smaller interactions or rounds, but these are treated as part of the game rather than as separate data entities.

All levels remain accessible regardless of whether previous levels have been completed. Children can earn journey points, receive in-game coins or cosmetic accessories, and use their rewards to customize a selected character.

Journey points accumulate across completed games and are displayed on the personal scoreboard. In game coins are tracked separately and may be spent on cosmetic accessories. Rewards are granted upon the first completion of each game, including when individual exercises have been skipped. A reward may include in-game coins, a cosmetic accessory, both, or neither.

7.1 Major Data Entities

The system manages the following major data entities:

- **Procedure Game:** Represents a game designed to familiarize children with a particular ophthalmology procedure, such as vision testing, eye-drop administration, or slit-lamp examinations. Each procedure has one corresponding game that represents a level in the child's journey. A game may contain multiple interactive exercises or rounds, and its content may vary according to the child's age group.
- **Character:** Represents a character available for the child to select and use throughout the preparation journey. Each character has a unique identifier, a display name, and a visual representation.
- **Cosmetic Accessory:** Represents a virtual item, such as a hat or glasses, used to customize a character's appearance. Accessories may be purchased using in-game coins or received directly as game completion rewards. Cosmetic accessories do not affect gameplay performance or clinical preparation activities.
- **Character Inventory:** Represents the cosmetic accessories owned by the child and identifies which accessories are currently equipped on the selected character. Owned items remain available when the child changes characters.
- **Reward Definition:** Represents the predefined reward associated with the first completion of a procedure game. Each reward includes journey points, and may also include in-game coins, a cosmetic accessory, both, or neither. Rewards are based on game completion rather than the accuracy of individual responses.
- **Local Preferences:** Represents locally stored application settings, including the child's optional nickname, selected character, age group, display or accessibility preferences, and caregiver acknowledgment of the application's educational purpose. These settings do not require a patient account.
- **Game Progress:** Represents the completion status, journey points earned, and reward claim status for each preparation game. Progress is tracked at the game level rather than for individual rounds or exercises. This information reflects participation in the application rather than clinical performance or assessment results.

- **Journey Progress:** Represents the child's overall progression across all preparation games, including accumulated journey points, completed levels, and the current in-game coin balance. Total journey points and completed levels are calculated from Game Progress records, while the spendable coin balance is maintained locally and updated when coins are earned or spent.

The relationship between these entities is as follows: each Procedure Game represents one ophthalmology procedure and corresponds to one level in the journey. Each game has an associated Reward Definition and a Game Progress record.

Local Preferences identifies the character selected by the child, while Character Inventory tracks the Cosmetic Accessories owned and equipped by the child.

When a game is completed for the first time, the system records its completion, awards the corresponding journey points, and grants any coins and cosmetic accessory in its Reward Definition. Game Progress records whether the reward has been claimed to prevent duplicate rewards when a completed game is replayed.

Journey Progress summarizes individual Game Progress records to calculate accumulated journey points and completed levels. The coin balance is maintained separately because coins may be spent in the character customization shop.

Section 6.1 specifies formally how level progress and rewards change during play.

7.2 Data Dictionary

The following data dictionary defines the major attributes associated with each entity.

Table 5: Data Dictionary for Journey-Based Progression

Entity	Attribute	Type	Description
Procedure Game	game_id	String	Unique identifier for a preparation game.
Procedure Game	procedure_name	String	Name of the ophthalmology procedure represented by the game.
Procedure Game	game_name	String	Name displayed for the preparation game.
Procedure Game	game_description	String	Clinically reviewed educational description of the procedure and its corresponding game.
Procedure Game	level_number	Integer	Level number representing the game's position in the suggested journey order.
Procedure Game	supported_age_groups	Set of Strings	Age groups for which the game provides appropriate content.
Character	character_id	String	Unique identifier for an available character.
Character	character_name	String	Name displayed for the character.

Continued on next page

Table 5 – continued from previous page

Entity	Attribute	Type	Description
Character	character_visual	String	Identifier of the character's visual representation within the application.
Cosmetic Accessory	accessory_id	String	Unique identifier for a cosmetic accessory.
Cosmetic Accessory	accessory_name	String	Name displayed for the accessory.
Cosmetic Accessory	accessory_visual	String	Identifier of the accessory's visual representation within the application.
Cosmetic Accessory	accessory_category	String	Category indicating where the accessory can be equipped, such as hats or glasses.
Cosmetic Accessory	coin_price	Optional Non-negative Integer	Number of in-game coins required to purchase the accessory. An absent price indicates that the item is not available for purchase.
Character Inventory	owned_accessory_ids	Set of Strings	Identifiers of accessories purchased with coins or received directly as game rewards.
Character Inventory	equipped_accessory_ids	Set of Strings	Identifiers of owned accessories currently equipped on the selected character.
Reward Definition	game_id	String	Identifier of the procedure game associated with the first-completion reward.
Reward Definition	points_awarded	Non-negative Integer	Journey points awarded when the game is completed for the first time.
Reward Definition	coin_amount	Non-negative Integer	Number of coins awarded. A value of 0 means the reward includes no coins.
Reward Definition	reward_accessory_id	Optional String	Identifier of the cosmetic accessory awarded. An absent value means the reward includes no accessory.

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Table 5 – continued from previous page

Entity	Attribute	Type	Description
Local Preferences	nickname	Optional String	User-defined text used to personalize the application. The nickname does not need to represent the child's actual name.
Local Preferences	selected_character_id	Optional String	Identifier of the character selected by the child for the preparation journey.
Local Preferences	age_group	Optional Enumeration	Age group used to choose the content of each level (FR7). Values: 3 to 4 years, 5 to 7 years.
Local Preferences	display_settings	Settings Record	Narration volume, music and sound-effect volume, visual cue size, and reduced motion (FR41).
Local Preferences	notice_acknowledged	Boolean	Indicates whether the caregiver has acknowledged the application's educational-use notice.
Game Progress	game_id	String	Identifier of the game associated with the progress record.
Game Progress	completion_status	Enumeration	Current completion status of the game. Values: Not Started, In Progress, Completed.
Game Progress	points_earned	Non-negative Integer	Journey points earned through the game's first completion. Points are retained when the game is replayed.
Game Progress	reward_claimed	Boolean	Indicates whether the game's first-completion reward has already been granted.
Journey Progress	total_points	Derived Integer	Total journey points accumulated across all preparation games, calculated from Game Progress records.
Journey Progress	completed_levels	Derived Integer	Number of preparation games marked as Completed, calculated from Game Progress records.
Journey Progress	total_levels	Derived Integer	Total number of preparation games available within the journey.

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Table 5 – continued from previous page

Entity	Attribute	Type	Description
Journey Progress	coin_balance	Non-negative Integer	Current spendable balance of in-game coins. The balance increases when coins are earned and decreases when accessories are purchased.

7.3 Privacy and Data Retention

EyeCanDo follows a data-minimization approach, collecting and retaining only the information necessary to support character customization, journey progression, rewards, user preferences, and local game progress. The application's privacy and data management practices are outlined below:

- **Data Collection:** The application does not require patient accounts, medical records, appointment details, or personally identifying information. Optional nicknames are used solely for personalization and do not need to correspond to the child's legal name.
- **Character Selection and Customization:** The application stores the selected character, owned cosmetic accessories, and equipped accessories locally. Character customization does not require registration or the collection of personal information.
- **Game and Journey Progress:** The application stores completion status, journey points, and reward claim status for each preparation game. Total points and completed levels are calculated from these records and displayed on the personal scoreboard. This information does not represent clinical performance or visual ability.
- **Rewards and Virtual Currency:** In-game coins and cosmetic accessories are earned through game completion. Coins may be exchanged for accessories within the application, while some accessories are awarded directly. Coin balances and owned accessories are stored locally. No real-money purchases or external payment services are required.
- **Data Storage and Protection:** User preferences, character selections, game progress, coin balances, and accessory inventories are stored locally on the user's device and protected from access by other applications (NFR26). The application does not transmit this information to external servers or require user accounts (NFR28, C7).
- **Data Retention and Deletion:** Locally stored information is retained until deleted by the caregiver or through application removal, subject to the device's backup and restoration behaviour. Caregivers can clear stored preferences, character selections, game progress, points, coin balances, and accessory inventories through the application's data management functionality.
- **Consent and Access:** The application does not require registration or the collection of personal information to access its preparation games. Caregivers acknowledge the application's educational purpose during initial setup. All games remain available regardless of whether previous levels have been completed.
- **Scoreboard Privacy:** The personal scoreboard displays only the child's accumulated journey points, completed levels, and current coin balance. It does not compare scores with other children or require an online leaderboard.
- **Offline Functionality:** Procedure games, characters, cosmetic accessories, and reward definitions are available locally. Game progress, rewards, accessory purchases, and character customization do not require an Internet connection or external server.

- **Research and Regulatory Compliance:** Any personal information collected separately for research or usability evaluation is outside the application's operational data storage and is subject to applicable institutional ethics and privacy requirements.

8 Non-Functional Requirements

The following non-functional requirements define the quality attributes and constraints of the application. Each requirement includes a rationale and a measurable fit criterion. Each requirement may also include a *Traces* line that lists the functional requirements that help satisfy it. Each requirement has a *Stakeholder Requirements* line that lists the stakeholder requirements it satisfies. A requirement that comes from the Hazard Analysis also has a *Hazard Analysis* line, in the same form as in Section 6.2.

8.1 Look and Feel Requirements

NFR1: Interactive controls that serve the same function shall remain in consistent screen positions across all screens where they appear.

Rationale: Consistent control placement helps children locate and recognize controls without relying on text labels. Maintaining consistent positions across different activities and game features reduces confusion and supports independent navigation.

Fit Criterion: A design review confirms that controls serving the same function remain within `CONTROL_POSITION_TOLERANCE_MM` millimetres of their designated positions across all applicable screens.

Stakeholder Requirements: STR3, STR18.

NFR2: No image, animation, narration, or sound shall be frightening, or show pain or distress beyond the sensations in the approved script.

Rationale: The clinic is already unfamiliar to the child, so the application must not add to the fear it is meant to reduce. Child-life specialists know what frightens children of this age, and the team does not.

Fit Criterion: A child-life specialist reviews all content against a checklist, and no finding is unresolved at release.

Traces: FR8.

Hazard Analysis: SR-8 (H2-6), SR-9 (H2-7), SR-23 (H2-21).

Stakeholder Requirements: STR2, STR6, STR15.

8.2 Usability and Humanity Requirements

NFR3: Every interactive element shall have a touch target at least `TOUCH_TARGET_MIN` millimetres wide and high on screen.

Rationale: The Apple and Android guidance minimums are about 7 mm and 9 mm [11, 12]. Children aged 3 to 7 have less precise motor control, so a larger size is used. The value is to be confirmed in usability testing.

Fit Criterion: Measured on each reference device, every interactive element meets this size.

Hazard Analysis: SR-24 (H2-3).

Stakeholder Requirements: STR3, STR17.

NFR4: Leaving a level for the journey map (FR20, FR24) shall take one touch from every screen of the level.

Rationale: A child or parent who chooses the wrong activity, or wants to stop, needs a way out that does not depend on reading. Without one, the child may feel trapped, which works against the goal of reassurance.

Fit Criterion: On every screen of every level, a tester reaches the journey map with one touch.

Traces: FR20, FR21, FR24.

Hazard Analysis: SR-4 (H2-2), SR-13 (H2-13).

Stakeholder Requirements: STR2.

NFR5: A first-time user from the defined user characteristics shall be able to complete each core task without reading instructions or receiving prior training. The core tasks are starting a chosen level from the journey map (UC3), playing a level to its end (UC5), and leaving a level early (UC8).

Rationale: The application is intended for use at home with little or no assistance. Users should be able to understand and interact with its core features independently, supporting engagement and reducing reliance on parental guidance.

Fit Criterion: In usability testing, at least `UNAIDED_RATE_MIN`% of participants complete each core task without verbal assistance, with a median completion time per task of at most `CORE_TASK_MAX_MIN` minutes. Participants aged 3–4 may have a parent present who does not explain or guide the task.

Traces: FR3, FR12, FR13.

Hazard Analysis: SR-24 (H2-3).

Stakeholder Requirements: STR3, STR4, STR18.

NFR6: A clinician who has not used the application before shall be able to open the introduction of a named level within `CLINICIAN_FIND_MAX_MIN` minute of picking up the device, after at most `CLINICIAN_TRAINING_MAX_MIN` minutes of training.

Rationale: Clinicians have little time per patient and little time for training (Section 2.4). A clinician who cannot quickly find the relevant level will not demonstrate it during a consultation, which is one of the frustrations of the clinician persona (Section 4).

Fit Criterion: In testing with clinicians or clinical staff, every participant meets both limits.

Traces: FR3, FR4.

Stakeholder Requirements: STR7.

NFR7: The application shall not use designs that pressure a child to keep playing or to come back. It shall have no rewards for playing on consecutive days, no rewards or offers that expire, no countdown timers outside an exercise, no randomized rewards, and no notifications.

Rationale: Young children cannot recognize these designs, and app store policies for children's applications restrict them (STR16). The rewards exist to encourage preparation, not time spent in the application.

Fit Criterion: A design review against a checklist of these patterns finds none, and the application does not request permission to send notifications.

Traces: FR28, FR29.

Stakeholder Requirements: STR8, STR16.

NFR8: Every narrated sentence shall have at most SENTENCE_MAX_WORDS words and use words familiar to a child aged 3 to 7, and text in the caregiver area shall be at or below Grade ADULT_TEXT_MAX_GRADE reading level.

Rationale: Young children follow short sentences in familiar words, and long ones are lost. Adults vary in reading ability, and Grade ADULT_TEXT_MAX_GRADE matches the defined user characteristics. The sentence limit is to be confirmed with the child-life specialists.

Fit Criterion: A count over all scripts finds no narrated sentence over SENTENCE_MAX_WORDS words, a child-life specialist finds no unfamiliar word, and a readability formula gives a grade of at most ADULT_TEXT_MAX_GRADE for each caregiver-area text.

Traces: FR12, FR39.

Hazard Analysis: SR-9 (H2-7), SR-24 (H2-3).

Stakeholder Requirements: STR1, STR3, STR18.

NFR9: The application shall meet the WCAG 2.2 Level AA success criteria that apply to a native mobile application.

Rationale: Children who attend an eye clinic may have low vision, and some will have hearing or motor differences. WCAG is the accepted benchmark [6], and the same duty applies in Ontario [7].

Fit Criterion: An accessibility audit against the applicable criteria finds no failure.

Traces: FR12, FR14, FR15, FR16, FR41.

Hazard Analysis: SR-16 (H2-10), SR-18 (H2-14), SR-19 (H2-15), SR-24 (H2-3).

Stakeholder Requirements: STR17.

NFR10: Every state and prompt shall remain distinguishable without colour, including for users with common forms of colour vision deficiency. A correct answer shown by FR16 shall be marked with at least two cues that do not rely on colour, such as an outline, an icon, or motion.

Rationale: Colour vision deficiency is common enough that some children and parents will have it, and it is more likely among users of an eye service. Relying on colour alone, for example green for correct and red for wrong, would hide information from them. A revealed answer needs two such cues because it must stand out from the other choices for a child who can neither read nor tell the colours apart.

Fit Criterion: Screens viewed under protanopia, deuteranopia, and tritanopia simulation show every state and choice as distinct by shape, position, or text as well as colour, and every revealed answer carries at least two cues that remain visible in all three simulations.

Traces: FR3, FR16.

Hazard Analysis: SR-16 (H2-10), SR-24 (H2-3).

Stakeholder Requirements: STR17.

8.3 Performance Requirements

NFR11: The home screen shall appear within STARTUP_MAX_S seconds of launch.

Rationale: A child looking at a blank screen may lose interest before anything starts. Low-spec devices set the limit, and the value is revisited after proof-of-concept measurements.

Fit Criterion: The median of `STARTUP_TRIALS_MIN` cold starts on each low-spec device is at most `STARTUP_MAX_S` seconds.

Stakeholder Requirements: STR3, STR10.

NFR12: The system shall respond visibly to a touch on an interactive element within `RESPONSE_MAX_MS` milliseconds. After an incorrect answer, the correct answer (FR16) shall appear within `ANSWER_REVEAL_MAX_S` second.

Rationale: About 0.1 seconds is the longest delay that feels instant to a user [13]. A child who sees no reaction will tap again, which the activities then have to handle. A correct answer that appears too late is no longer linked to the question in the child's mind.

Fit Criterion: At least `RESPONSE_RATE_MIN`% of measured touches on each low-spec device receive a visible response within `RESPONSE_MAX_MS` milliseconds, and every measured answer reveal on each low-spec device appears within `ANSWER_REVEAL_MAX_S` second.

Traces: FR16, FR19.

Hazard Analysis: SR-2 (H1-2), SR-3 (H2-1), SR-16 (H2-10).

Stakeholder Requirements: STR2.

NFR13: Animation shall run at no fewer than `FRAME_RATE_MIN` frames per second.

Rationale: Choppy animation of a procedure would distract from what it shows.

Fit Criterion: On each low-spec device, the frame rate is at least `FRAME_RATE_MIN` frames per second for at least `FRAME_RATE_COMPLIANCE_MIN`% of the time over a full walkthrough and practice activity.

Hazard Analysis: SR-2 (H1-2).

Stakeholder Requirements: STR2, STR3.

NFR14: Narration shall start within `AV_SYNC_MAX_MS` milliseconds of the animation it accompanies.

Rationale: Narration that does not match the picture can teach the wrong step. The value is a practical limit that keeps the two looking together, and it is checked in the proof of concept.

Fit Criterion: Measured on each reference device across all steps, no offset exceeds this value.

Traces: FR12.

Hazard Analysis: SR-25 (H2-17).

Stakeholder Requirements: STR1.

NFR15: No sound effect or music shall have a peak level above the peak level of the narration.

Rationale: A sudden loud sound can frighten a child who is already anxious, and can be uncomfortable for small ears. Tying every sound to the narration level, which the parent sets with the device volume, avoids a sound that is louder than the voice.

Fit Criterion: An audio analysis of all sound assets finds no exception.

Traces: FR15.

Hazard Analysis: SR-14 (H2-16).

Stakeholder Requirements: STR2, STR15.

NFR16: No content shall flash more than `FLASH_MAX_PER_S` times in any one-second period.

Rationale: Flashing content can trigger seizures in people with photosensitive epilepsy. This is the limit of WCAG 2.2 success criterion 2.3.1, Three Flashes or Below Threshold [6]. It is applied to all content, without the exception for small flashing areas, because a young child may hold the device close to their face.

Fit Criterion: Every animation and video is checked with a photosensitive-epilepsy analysis tool, and no sequence exceeds the limit.

Hazard Analysis: SR-14 (H2-16).

Stakeholder Requirements: STR15.

NFR17: At least `CRASH_FREE_RATE_MIN`% of sessions shall end without a crash or freeze.

Rationale: A crash in front of an anxious child, or in a waiting room minutes before an appointment, harms trust in the application. Testing cannot show that an application never crashes, so the requirement sets a rate that testing can measure.

Fit Criterion: At least `CRASH_FREE_RATE_MIN`% of at least `TEST_SESSIONS_MIN` test sessions end without a crash or freeze.

Traces: FR24, FR25, FR26, FR27.

Hazard Analysis: SR-4 (H2-2), SR-20 (H2-18), SR-22 (H2-20).

Stakeholder Requirements: STR2.

NFR18: The application shall keep its CPU utilization at or below `CPU_UTILIZATION_MAX_PERCENT`% during gameplay on supported low-spec devices.

Rationale: Young children may hold a device for long periods, and excessive hardware utilization can contribute to device heating, causing discomfort or minor injury. Limiting CPU utilization provides a software-verifiable way to control sustained hardware load. The utilization limit will be revisited after proof-of-concept measurements on low-spec devices to ensure it is practical and helps prevent excessive heating.

Fit Criterion: On each low-spec device, CPU utilization is measured throughout a continuous session of at least `HEAT_TEST_SESSION_MIN` minutes of gameplay. The measured CPU utilization shall remain at or below `CPU_UTILIZATION_MAX_PERCENT`% throughout the session.

Hazard Analysis: SR-1 (H1-1).

Stakeholder Requirements: STR15.

8.4 Operational and Environmental Requirements

NFR19: After installation, every function shall work with no network connection.

Rationale: Hospital waiting rooms often have poor connectivity, and that is a place where the application is useful. A network dependency would also raise privacy questions. This requirement

makes C5 measurable.

Fit Criterion: The full functional test suite passes with the device in airplane mode.

Traces: FR26, FR28, FR34.

Hazard Analysis: SR-26 (H2-4).

Stakeholder Requirements: STR4.

NFR20: A parent shall be able to install the application from the app store and reach the home screen without creating an account, granting permissions, or downloading further content.

Rationale: Every extra step is a place where a parent may give up, and a required account would conflict with the decision to collect no personal data.

Fit Criterion: On each reference device, a tester installs the application and reaches the home screen following only the store listing, with none of those steps required.

Hazard Analysis: SR-26 (H2-4).

Stakeholder Requirements: STR4.

NFR21: The installed size of the application shall be at most `INSTALL_SIZE_MAX_MB` megabytes.

Rationale: The limit is chosen so a family can download over a mobile connection and so older devices with little storage can hold it. The proof of concept measures the real size.

Fit Criterion: The size reported by each app store for the release build is within this limit.

Stakeholder Requirements: STR4, STR10.

NFR22: Every level and every entry of the procedure guide shall be approved by a paediatric ophthalmologist, as matching the procedure at the clinic, before it is released. Content that changes after approval shall be approved again.

Rationale: Children and caregivers rely on the game to show what will really happen (STR5), and the team has no clinical training. A record of approval for each version shows that the released content is the content that was approved. NFR27 then makes sure it is not changed on the way to the device.

Fit Criterion: For each release, an approval record signed by a paediatric ophthalmologist lists every level and guide entry at its released version, with no finding unresolved.

Traces: FR6, FR8, FR9, FR39.

Hazard Analysis: SR-5 (H2-5), SR-7 (H2-5).

Stakeholder Requirements: STR1, STR5, STR6.

8.5 Maintainability and Support Requirements

NFR23: The effort to make any likely change in the Likely Changes section shall be less than `MAINT_FRACTION` of the original development time, not counting clinical review.

Rationale: A change such as a new procedure should cost far less than building the first one, or the application cannot keep up with what clinicians want covered.

Fit Criterion: The recorded effort for each such change made during the project is within this

fraction.

Stakeholder Requirements: STR11.

NFR24: The application shall run on iOS `MIN_IOS` and later and Android `MIN_ANDROID` and later, on phones and tablets with screens from `SCREEN_MIN_IN` to `SCREEN_MAX_IN` inches.

Rationale: Families use the devices they already own. These versions cover the large majority of devices in use by families and keep the supported range testable by a small team. The performance requirements are measured on low-spec devices, so that their limits hold for the oldest devices that families still use. The values are revisited after the proof of concept. This requirement makes C1 measurable.

Fit Criterion: The functional tests in the VnV Plan pass on every reference device.

Stakeholder Requirements: STR10.

NFR25: Every third-party library and engine component shall be fixed to an exact version, and each change to one shall be tested on a release build before it is included in a release.

Rationale: The application depends on software the team does not control, and an unexpected update or incompatibility could cause crashes or wrong behaviour in the activities that prepare the child. Fixing versions means each release contains only what was tested.

Fit Criterion: The dependency list gives an exact version for every third-party library and engine component, and the functional test suite passes on the release build after each dependency change, with no finding unresolved at release.

Hazard Analysis: SR-22 (H2-20).

Stakeholder Requirements: STR2.

8.6 Security Requirements

NFR26: Stored settings and progress shall be readable only by the application, using the storage the platform keeps private to it.

Rationale: The nickname and progress are about a child, and no other application or person using the device should read them. The platform already provides private storage.

Fit Criterion: On an unmodified test device, no other application and no file-sharing function can read the stored data.

Traces: FR26, FR40.

Hazard Analysis: SR-21 (H2-19).

Stakeholder Requirements: STR13.

NFR27: Every release build shall be digitally signed and distributed only through the Apple App Store and Google Play Store.

Rationale: What children see must be what clinicians approved. A signed build from the official stores means the content cannot have been altered on the way to the device.

Fit Criterion: The signature of each release build verifies with the team's signing key using the platform tools, and no other distribution channel is in use.

Hazard Analysis: SR-5 (H2-5), SR-7 (H2-5).

Stakeholder Requirements: STR5, STR16.

NFR28: The application shall not transmit any data off the device.

Rationale: Not collecting data removes most of the privacy and legal exposure for a service aimed at children, who cannot consent, and the application does not need any data to work. Evaluation measures are collected separately, with consent. This requirement makes C6 measurable.

Fit Criterion: A network capture during a full test session shows no outbound request from the application.

Hazard Analysis: SR-21 (H2-22), SR-22 (H2-22).

Stakeholder Requirements: STR13.

NFR29: The application shall store only the attributes of the Local Preferences, Character Inventory, Game Progress, and Journey Progress entities in the data dictionary (Table 5).

Rationale: Data that is not stored cannot leak or be requested. Only these items are needed for the application to function. This requirement makes C7 measurable.

Fit Criterion: Inspection of the storage after a full test session finds no other data.

Traces: FR40, FR42, FR43.

Hazard Analysis: SR-21 (H2-19).

Stakeholder Requirements: STR13.

NFR30: Each release shall contain no third-party library with a publicly known vulnerability of high or critical severity.

Rationale: The application has no server and accepts no input except touch, so its main exposure is through the libraries it includes. A compromised dependency could break the privacy requirements.

Fit Criterion: A dependency vulnerability scan run on each release build reports none of that severity.

Hazard Analysis: SR-22 (H2-22).

Stakeholder Requirements: STR13.

8.7 Cultural and Political Requirements

NFR31: Content shall not assume a child's family structure, culture, religion, gender, or ethnicity.

Rationale: Families using the service come from many backgrounds, and a child who does not see themselves in the content, or is shown a family unlike theirs, may feel left out when the aim is reassurance.

Fit Criterion: The review of child-facing content includes an inclusivity checklist, and no finding is unresolved at release.

Traces: FR32.

Hazard Analysis: SR-15 (H2-9).

Stakeholder Requirements: STR12.

8.8 Compliance Requirements

NFR32: The application, its notices, and its store listings shall make no claim to diagnose, screen, treat, or monitor any condition.

Rationale: Software that makes such claims may be regulated as a medical device [8]. Such a claim would also encourage a parent to treat the practice as a vision test and miss a real problem.

Fit Criterion: Two team members and a clinician review every child-facing text, adult-facing text, and store listing against a checklist, and find no such claim.

Traces: FR1, FR39.

Hazard Analysis: SR-10 (H2-8).

Stakeholder Requirements: STR14.

NFR33: Every third-party asset and library shall have a recorded licence that permits redistribution under the project's chosen software licence.

Rationale: Assets from a public store may forbid redistribution or require terms that conflict with the project's chosen licence. A removal for a licence breach would take the application away from families. This requirement makes C2 measurable.

Fit Criterion: The licence log lists every third-party asset and library with its licence, and has no gap.

Hazard Analysis: SR-23 (H2-21).

Stakeholder Requirements: STR16.

NFR34: The application shall meet the policies for children's applications of both app stores: the Kids Category rules of the App Store Review Guidelines [9] and the Families policy of Google Play [10].

Rationale: The application is aimed at children, and the policies call for parental gates and limit advertising, outside links, and third-party data collection [9, 10]. Non-compliance would lead to rejection or removal. This requirement makes the children's-application part of C8 measurable.

Fit Criterion: Both app stores approve the release build for a children's audience, and every item on the pre-submission policy checklist passes.

Traces: FR37, FR38.

Stakeholder Requirements: STR15, STR16.

9 Requirements Traceability Matrices

Table 6 traces each use case to the stakeholder requirements (Section 3) that it addresses and to the functional requirements that refine it.

Table 6: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Stakeholder Requirements	Use Case	Functional Requirements
STR4, STR10, STR13, STR14	UC1	FR1, FR2
STR4, STR12, STR18	UC2	FR31, FR32
STR1, STR3, STR4, STR7, STR8, STR9, STR11, STR16, STR18	UC3	FR3, FR4, FR5
STR1, STR5, STR6, STR7, STR11, STR14	UC4	FR1, FR39
STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR15, STR17, STR18	UC5	FR6, FR8, FR9, FR11, FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19
STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR15, STR17, STR18	UC6	FR6, FR8, FR9, FR10, FR11, FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19
STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR15, STR17, STR18	UC7	FR6, FR8, FR9, FR11, FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19
STR2, STR7, STR9	UC8	FR20, FR21, FR22, FR23, FR24, FR25, FR26
STR3, STR8, STR9, STR16	UC9	FR5, FR28, FR29
STR3, STR8, STR9, STR16	UC10	FR30
STR3, STR8, STR12, STR15, STR16	UC11	FR32, FR33, FR34, FR35, FR37
STR3, STR8, STR12	UC12	FR32, FR36
STR3, STR13, STR16, STR17	UC13	FR7, FR14, FR38, FR40, FR41
STR13, STR16	UC14	FR38, FR42, FR43
STR4, STR10, STR13, STR15	UC15	FR26, FR27

Table 7 traces each stakeholder requirement to the non-functional requirements and the system constraints that help satisfy it. Together with Table 6, it shows that every stakeholder requirement is refined by at least one use case or non-functional requirement.

Table 7: Traceability Between Stakeholder Requirements, Non-Functional Requirements, and Constraints

Stakeholder Req	Non-Functional Requirements	Constraints
STR1	NFR8, NFR14, NFR22	–
STR2	NFR2, NFR4, NFR12, NFR13, NFR15, NFR17, NFR25	–
STR3	NFR1, NFR3, NFR5, NFR8, NFR11, NFR13	–
STR4	NFR5, NFR19, NFR20, NFR21	C5
STR5	NFR22, NFR27	–
STR6	NFR2, NFR22	–
STR7	NFR6	–
STR8	NFR7	–
STR9	–	–
STR10	NFR11, NFR21, NFR24	C1, C4
STR11	NFR23	–
STR12	NFR31	–
STR13	NFR26, NFR28, NFR29, NFR30	C6, C7
STR14	NFR32	–
STR15	NFR2, NFR15, NFR16, NFR18, NFR34	C8
STR16	NFR7, NFR27, NFR33, NFR34	C2, C3, C8
STR17	NFR3, NFR9, NFR10	–
STR18	NFR1, NFR5, NFR8	–

Table 8 lists the functional requirements that help satisfy a non-functional requirement. Functional requirements without a related non-functional requirement are omitted.

Table 8: Traceability Between Functional and Nonfunctional Requirements

Functional Req	Nonfunctional Req
FR1	NFR32
FR3	NFR5, NFR6, NFR10
FR4	NFR6
FR6	NFR22
FR8	NFR2, NFR22
FR9	NFR22
FR12	NFR5, NFR8, NFR9, NFR14
FR13	NFR5
FR14	NFR9

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Table 8 – continued from previous page

Functional Req	Nonfunctional Req
FR15	NFR9, NFR15
FR16	NFR9, NFR10, NFR12
FR19	NFR12
FR20	NFR4
FR21	NFR4
FR24	NFR4, NFR17
FR25	NFR17
FR26	NFR17, NFR19, NFR26
FR27	NFR17
FR28	NFR7, NFR19
FR29	NFR7
FR32	NFR31
FR34	NFR19
FR37	NFR34
FR38	NFR34
FR39	NFR8, NFR22, NFR32
FR40	NFR26, NFR29
FR41	NFR9
FR42	NFR29
FR43	NFR29

Table 9 traces the safety and security requirements of the Hazard Analysis to the functional requirements and the non-functional requirements of Section 8 that satisfy them. The numbers of the safety and security requirements (SR) and of the hazards (H) are those of the Hazard Analysis.

Table 9: Traceability Between the Hazard Analysis and Functional and Non-Functional Requirements

SR	Hazard	Safety and Security Req	Functional Req	Non-Functional Req
SR-1	H1-1	Device heat limit	–	NFR18
SR-2	H1-2	Device performance	–	NFR12, NFR13
SR-3	H2-1	Prevention of unintentional gameplay actions	FR19	NFR12
SR-4	H2-2	Gameplay recovery	FR20, FR22, FR23, FR24	NFR4, NFR17

Continued on next page

Table 9 – continued from previous page

SR	Hazard	Safety and Security Req	Functional Req	Non-Functional Req
SR-5	H2-5	Clinical approval of procedure content	–	NFR22, NFR27
SR-6	H2-5	Variation between clinics	FR9	–
SR-7	H2-5	Release of approved content only	–	NFR22, NFR27
SR-8	H2-6	Honest description of sensations	FR8	NFR2
SR-9	H2-6, H2-7	Child-life review of content	–	NFR2, NFR8
SR-10	H2-8	Medical disclaimer and caregiver acknowledgment	FR1, FR2	NFR32
SR-11	H2-11	All preparation levels remain accessible	FR4	–
SR-12	H2-12	Encouraging feedback without penalties	FR13, FR18, FR28	–
SR-13	H2-13	Termination of audio and visual effects on exit	FR21	NFR4
SR-14	H2-16	Safe audio and visual effects	FR15, FR41	NFR15, NFR16
SR-15	H2-9	Diverse and inclusive characters and rewards	FR32	NFR31
SR-16	H2-10	Revealing correct answers	FR16	NFR9, NFR10, NFR12
SR-17	H2-10	At least one correct answer	FR17	–
SR-18	H2-14	Guidance for muted narration	FR15, FR25	NFR9
SR-19	H2-15	Visual cues during gameplay	FR14, FR38, FR41	NFR9
SR-20	H2-18	Reliable save data handling	FR26, FR27	NFR17
SR-21	H2-19, H2-22	Minimization and protection of game data	–	NFR26, NFR28, NFR29
SR-22	H2-20, H2-22	Third-party software validation	–	NFR17, NFR25, NFR28, NFR30
SR-23	H2-21	Third-party asset review	–	NFR2, NFR33

Continued on next page

Table 9 – continued from previous page

SR	Hazard	Safety and Security Req	Functional Req	Non-Functional Req
SR-24	H2-3	Accessible, age-appropriate interaction	FR7, FR12, FR41	NFR3, NFR5, NFR8, NFR9, NFR10
SR-25	H2-17	Consistent narration and animation	FR25	NFR14
SR-26	H2-4	Offline availability of preparation content	–	NFR19, NFR20

10 Rationale

10.1 Scope

The scope is focused on preparing children aged 3 to 7 for common paediatric ophthalmology procedures. Diagnosis, treatment, and other clinical functions were excluded because they are outside the main goal of the application and would add requirements that are not needed for preparation. The initial set of procedures is also limited so that the team can focus on making each activity clear, accessible, and reliable. Features such as AR and personalized preparation are considered stretch goals rather than part of the minimum scope.

10.2 Functional Requirements

Each functional requirement has its own rationale in Section 6.2. This subsection explains only the decisions that shape most of the requirements.

The requirements follow the journey design recommended by the project supervisor (see the Data Requirements rationale below). The procedures needed at an appointment can change at short notice, so every level is open at all times and progress is recorded per level. The journey map, the rewards, the scoreboard, and the characters all build on this structure.

The format of each level is not yet decided. The requirements therefore state what every level must convey and how the journey records progress, but not what happens inside a level. Requirements for the inside of each level will be added once the format is decided.

The application must not become a new source of stress for a child who is already anxious. The application therefore has no failure states. A wrong action leads to a hint, a wrong answer leads to the correct answer being shown, exercises can be skipped, rewards do not depend on answers, and the child can leave at any time without losing anything. Wherever a requirement could cost the child progress, it is written so that it does not.

The users are children aged 3 to 7, most of whom cannot read. Every instruction is therefore given through narration and pictures, and nothing in the child area depends on reading. Everything that is not play, such as the medical notice, the settings, and deleting data, is placed in the caregiver area.

The application works offline and stores only the player state on the device. No account, network, or personal information is needed. The requirements on saving, restoring, and resetting data all follow from this decision.

The Hazard Analysis states its safety and security requirements at a general level. This SRS states them as specific, testable requirements. Those that describe what the system does are functional requirements, and each one names the safety and security requirement and the hazard it addresses. Those that describe a quality of the system or the release process are non-functional requirements.

The journey and the player's progress are specified formally in Section 6.1. They involve state over time, such as rewards given, levels started, and coins earned. Informal wording such as "the reward is given once" is easy to misread for this kind of behaviour. The player state is specified with discrete mathematics because it is a small set of values with clear operations. The status of each level is also shown as a state machine (T2), because it is easiest to check as a picture. Every requirement can then cite exactly the part of the model it depends on.

10.3 Data Requirements

The data requirements were developed by considering how ophthalmology procedures and their preparation games should be represented. Initially, the team considered a one-to-many relationship where each procedure could have multiple preparation activities. However, since the team had not established whether multiple distinct games would be developed for each procedure, a simpler one-to-one relationship was selected. Each procedure is represented by one game, which may contain multiple smaller exercises or rounds.

The team also considered two approaches for selecting preparation games: allowing caregivers to manually choose games based on clinical recommendations, or having caregivers enter a clinician-provided code to automatically select the appropriate games. The team consulted the project supervisor to determine which approach would be more practical in a clinical setting.

The supervisor recommended making all games accessible, since the procedures required during an examination may change unexpectedly. The supervisor also suggested presenting the games as a journey with levels, rewards, a personal scoreboard, and character customization.

Based on this feedback, the team removed the caregiver configuration and recommendation code entities from the data requirements. Instead, entities supporting character selection, game progress, journey progression, and rewards were introduced. All user data is stored locally to maintain offline functionality and minimize the collection of personal information.

Rewards are defined flexibly. A level reward may include coins, a cosmetic accessory, both, or neither, so that the team and the supervisors can tune the rewards without changing the data model.

10.4 Non-Functional Requirements

This section explains decisions that affect multiple non-functional requirements. Individual requirement rationales are provided alongside their respective requirements.

The main non-functional requirements support three priorities: child-friendly interaction, reliable use, and privacy. Since the application is designed for children aged 3 to 7, requirements such as NFR1, NFR3, and NFR5 ensure that activities are easy to navigate and complete. NFR2 and NFR8 ensure that the content is appropriate for the target age group, and NFR22 ensures that it matches what happens at the clinic. NFR7 keeps the rewards from pressuring the child to keep playing, and NFR6 lets a clinician demonstrate a level during a short consultation.

Accessibility is also important so that children with different needs can use the application. NFR9 and NFR10 address accessibility needs, while NFR14 and NFR15 help keep audio and animations clear during activities. NFR16 protects children with photosensitive epilepsy.

Since the application may be used at home or before an appointment, NFR19 allows activities to work without internet access. NFR12, NFR13, and NFR17 help prevent delays and interruptions that could affect the experience. NFR24 and NFR20 also make the application easier to access across supported devices.

Finally, privacy and security are especially important because the application is intended for children. NFR29 limits the information stored, NFR28 prevents data from being transmitted off the device, and NFR26 protects locally stored data. NFR27 and NFR30 help protect the application from unauthorized changes and vulnerable dependencies. Other requirements, including NFR31, NFR32, NFR33, and NFR34, address inclusive content, the application's medical scope, and distribution requirements.

11 Likely Changes

The following changes are expected during the project or after release. The design should make each of them easy to carry out (NFR23). Each change lists the parts of this document it affects.

LC1: The format of each level. The activities inside a level are not yet decided, and they will be chosen after prototyping and feedback from the supervisor.

Affects: FR6, Sections 6.2.3 and 6.2.4. The formal model (Section 6.1) is not affected because it does not describe the inside of a level.

LC2: The set of procedures. Levels may be added, removed, or reordered as the supervisor confirms which procedures are most common.

Affects: the game content (Section 6.1), the data dictionary (Table 5), the procedure guide (FR39), and the level use cases. The requirements and the formal model hold for any number of levels.

LC3: The procedure content. The wording, images, step order, and sensation descriptions of a level may change after clinical and child-life review.

Affects: the content of each level and of the procedure guide (FR8, FR39).

LC4: The rewards. The points, coins, and accessory given by each level, and the price of each accessory, will be tuned during testing.

Affects: the game content (Section 6.1) and the Reward Definition entity. A reward may already include coins, an accessory, both, or neither, so no requirement changes (FR28).

LC5: The characters, accessories, and accessory categories offered.

Affects: FR31, FR32, and FR33.

LC6: The values of the symbolic constants, after usability testing, performance measurements, and clinician feedback.

Affects: Table 20 only. Requirements refer to the constants by name.

LC7: Content for each age group. Levels may add separate content for younger and older children.

Affects: FR7 and the content of each level.

LC8: Additional languages, most likely French for families in Ontario.

Affects: the narration and text of every level, the procedure guide, and NFR8.

LC9: The supported operating system versions and screen sizes.

Affects: NFR24 and C1.

LC10: The safety and security requirements. The Hazard Analysis will be revised as the design develops, and this may add, remove, or renumber requirements.

Affects: the requirements that have a Hazard Analysis line, and Table 9.

LC11: The final art and audio, which will replace the placeholder assets used during development.

Affects: the content only. The assets must still meet NFR2 and NFR21.

LC12: Augmented reality presentation of procedures, which is a stretch goal. It would require access to the camera.

Affects: the level requirements, NFR20, and the Hazard Analysis.

LC13: Personalized preparation based on the child's appointment, which is a stretch goal.

Affects: the journey requirements (Section 6.2.2), the caregiver settings (Section 6.2.9), and the data requirements (Section 7).

LC14: The stakeholder requirements can expand once more interviews and elicitation are conducted with the primary stakeholders (children, caregivers) after ethics board approval. The stakeholder requirements will also be revised as the design develops, and this may add, remove, or renumber requirements.

Affects: Stakeholder Requirements in Section 3.

12 Unlikely Changes

The design can assume that the following will not change. Each one is stated with the reason it is considered stable.

ULC1: The primary users are children aged 3 to 7 who are preparing for a paediatric eye appointment, supported by a caregiver. This is the purpose of the project.

ULC2: EyeCanDo presents one journey with one level for each procedure, and every level is open at all times (FR4, T1). The supervisor recommended this design because the procedures needed at an appointment can change at short notice.

ULC3: EyeCanDo has no failure states. A wrong action leads to a hint, a wrong answer leads to the correct answer being shown, and exercises can be skipped without penalty (FR13, FR16, FR18). This follows from the goal of reducing anxiety.

ULC4: Each level reward is given once, when the level is finished for the first time (T3). The contents of a reward may change (LC4), but this rule will not.

ULC5: EyeCanDo works offline and sends no data off the device (C5, C6). Waiting rooms may have poor connectivity, and nothing needs to be transmitted.

ULC6: EyeCanDo stores no personal or health information (C7, NFR29). Changing this would change the privacy and ethical basis of the project.

ULC7: EyeCanDo runs on phones and tablets with touch screens, and it is built with a cross-platform engine (C1, C4). Families already own these devices, and dedicated hardware would defeat the goal of preparing at home.

ULC8: EyeCanDo is an educational preparation aid. It makes no diagnostic or treatment claim (NFR32, FR1). A medical function would make it a medical device, which is outside the scope of the project.

ULC9: EyeCanDo offers no purchases with real money and no advertising (FR37, C8). Both would be inappropriate for young children and conflict with app store policies for children's applications.

13 Appendices

13.1 Appendix A: Additional Materials

13.1.1 Requirements Elicitation Sources

Table 10 lists the sources from which the stakeholder requirements were elicited, with links to the GitHub issues that hold the meeting notes. The "Stakeholder Requirements" column gives the trace from each source to the stakeholder requirements in Section 3.

Table 10: Stakeholder Requirement Elicitation Sources

ID	Source and Date	Information Summary	Stakeholder Requirements
ER1	Supervisor's Project Description (September 2026)	Children often experience anxiety during ophthalmology visits, which can reduce cooperation and affect the quality of assessments. An interactive mobile application should prepare children for appointments through age-appropriate games, letting children practice at home and earn rewards, introducing common procedures, including vision testing, eye drops, and slit-lamp examinations. Designed alongside clinicians, child-life specialists, parents, and children, we will pilot and evaluate the app's usability, feasibility, anxiety reduction, cooperation, and clinic efficiency.	STR1, STR2, STR3, STR4, STR5, STR6, STR7, STR8
ER2	Supervisor Meeting 1 (2026/09/15) (#1)	Discussion of the age groups, whether difficulty should vary by age, and the structure of the game. The outcome is a target range of 3 to 7 years with augmented reality and personalization as stretch goals.	STR3, STR12, STR15, STR18
ER3	Stakeholder Elicitation in Problem Statement and Goals document (2026/09/28)	Ophthalmologists can evaluate the preparedness of the children through the successful completion of the games. Personalization to the child's visit is a stretch goal. Regulatory bodies require the design process and product to be safe and ethical for children; the app runs on iOS and Android phones and tablets.	STR7, STR9, STR10, STR15, STR16, STR17, STR18
ER4	Hazard Analysis Lecture (2026/10/05) (#136)	Users might treat the game as medical advice. This should be written as a hazard with the mitigation strategy as a "not official medical advice" disclaimer.	STR14

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Table 10 – continued from previous page

ID	Source and Date	Information Summary	Stakeholder Requirements
ER5	Supervisor Meeting 2 (Clinical Visit) (2026/10/07) (#138)	Procedures explained include: vision test with one eye covered, eye pressure, dilating eye drops, slit-lamp, headlamp examination, colour vision book, prism alignment test (essential before surgery), eye photos and OCT, glasses check, eye patching. Observed sequence with an adult: history of medical issue, vision test, pupil check with headlamp, eye drops, slit lamp, finger-following test, 20-minute wait for dilation, headlamp examination. Other notes include: the drops burn and are a major source of anxiety. Covering one eye is the best method for children to test one eye. The slit-lamp needs the head to stay very still. The supervisor also specified that an ethics board review will be needed for surveying and testing the project at the clinic.	STR6, STR11, STR17
ER6	Teaching Assistant Informal Meeting (2026/10/07) (#137)	Children should just play the game, but clinicians use their existing system to assign tests and host the personal information on their already secure system. Further elicitation is required from stakeholders to see if integration into the existing system is possible.	STR13
ER7	Literature Research	A study of 298 children undergoing cycloplegic eye examinations found that 26% experienced significant distress before receiving eye drops, while uncooperative children were also more anxious and required more time to complete the examination [5].	STR1
ER8	Supervisor Email Correspondence (2026/10/13)	The supervisor described a preferred design with three parts. Children have full access to all modules, because clinically they will do all of the scenarios, even if not on the same day. The games are presented as a journey with different "levels", with points and a scoreboard at the end. The child selects a unique character to play as.	STR7, STR8, STR12

13.2 Appendix B: Development Plan

This appendix identifies requirement priorities and six implementation phases for EyeCanDo, following the milestones in Section 9 of the Development Plan. Dates and coverage may be refined after the PoC, stakeholder feedback, and assessment of available resources. The tables distinguish planning, investigation, implementation, verification, and simulation.

Requirement priorities

Table 11 explains the highest-priority (P1), supporting (P2), and lower-priority (P3) requirements. Quality and safety safeguards apply whenever the associated features are developed. P3 applies only to the store-dependent and unavailable-device portions identified below.

Table 11: Requirement priorities and rationale

Priority	Linked requirements	Reason for priority
P1: Core	**Truthful and safe preparation.** FR1, FR2, FR6, FR8, FR9, FR10, FR37, NFR2, NFR7, NFR8, NFR16, NFR22, NFR31, NFR32	These requirements are essential because the application's main purpose is to prepare children for eye examinations. The content must be accurate, age-appropriate, and reassuring without providing misleading medical guidance or requiring real-money purchases.
P1: Core	**Understandable, forgiving interaction.** FR12, FR13, FR14, FR15, FR16, FR17, FR18, FR19, FR20, FR21, FR22, FR23, FR24, FR25, NFR3, NFR4, NFR5, NFR6, NFR9, NFR10, NFR15	Children aged three to seven need clear instructions and accessible controls to use the application. They must also be able to make mistakes, skip exercises, and exit activities without penalties or losing progress. These features are necessary for a safe and stress-free experience.
P1: Core	**Dependable access and caregiver control.** FR3, FR4, FR5, FR11, FR26, FR27, FR38, FR39, FR41, FR42, FR43, NFR17, NFR19, NFR26, NFR28, NFR29	Children must be able to access all preparation games without an Internet connection or completing earlier levels. Their progress must be saved reliably, while caregivers need control over application settings and stored information. These functions are essential for the application to work as intended.
P2: Supporting	**Progress and personalization.** FR7, FR28, FR29, FR30, FR31, FR32, FR33, FR34, FR35, FR36, FR40	These features make the application more engaging by allowing children to earn rewards, track their achievements, and customize their characters. They are developed after the core gameplay and progress-saving functions they depend on. Basic rewards are introduced in Phase 3, followed by additional customization features.

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Table 11 – continued from previous page

Priority	Linked requirements	Reason for priority
P3: Lower	**Actual publication and wider device coverage.** Publication-dependent parts of NFR20, NFR21, NFR27, NFR34; coverage beyond accessible devices in NFR24.	Actual app store publication requires external approval, while broader compatibility testing depends on access to additional devices. The team will prioritize completing and testing the application on available devices, simulate app store submission for the course, and attempt real publication if time permits.

Overall phase schedule

Table 12 summarizes the phase targets and their course milestones. The following phase tables provide the requirement-level detail for this schedule.

Table 12: Overall development phases and course milestones

Phase and target period	Phase objective	Milestone alignment
1: Foundations 19 Oct.–2 Nov. 2026	Identify major risks, prepare feasibility experiments, and plan the architecture and verification approach.	SRS and Hazard Analysis Rev. 0: 19 Oct. V&V Plan Rev. 0: 2 Nov.
2: Proof of concept 3 Nov.–4 Dec. 2026	Investigate content, child-interaction, and offline-media risks through independent experiments on accessible devices.	Design Rev. −1 and Fall CD: 16 Nov. PoC demonstration: by 4 Dec.
3: Core application 5 Dec. 2026– 18 Jan. 2027	Use PoC findings to implement setup, local progress, the preparation journey, basic rewards, and caregiver controls.	Design Rev. 0: 18 Jan.
4: Feature integration 19 Jan.–11 Feb. 2027	Integrate the scoreboard, characters, and accessory shop with saved progress and caregiver settings.	Revision 0 demonstration: by 11 Feb.
5: System verification 12 Feb.–8 Mar. 2027	Evaluate implemented features on accessible devices, address priority defects, and record remaining coverage gaps.	V&V Report Rev. 0 and Winter CD: 8 Mar.
6: Final delivery 9 Mar.–6 Apr. 2027	Finalize the demonstration and handover evidence, simulate store workflows, and attempt real publication if time permits.	Final demonstration: by 21 Mar. Final documentation: 31 Mar. Expo: 6 Apr.

Phase 1: Foundations

Identify the major risks related to educational content, user interactions, and media delivery, and prepare the experiments for the Proof of Concept (PoC). Arrange access to reviewers and establish the approval process for testing with children. Table 13 outlines the design and planning activities, while application implementation will begin after the PoC. The system constraints (C1–C8) will guide these decisions, taking into account the device and app store scope described below.

Table 13: Phase 1: Requirements addressed through planning

Requirement	Planned work
NFR19	Plan locally packaged content and offline operation.
NFR23	Plan reusable procedure, guidance, and data components.
NFR24	List accessible device models, OS versions, and screen sizes as the course test set.
NFR25	Select fixed dependency versions and define update checks.
NFR26	Choose private local storage for settings and progress.
NFR27	Identify development signing needs and the intended store workflow.
NFR28	Exclude network communication from the prototypes and application design.
NFR29	Define the permitted local data and retention rules.
NFR30	Plan dependency checks and review included libraries.
NFR32	Plan content review to keep the educational purpose clear.
NFR33	Start a licence and attribution record.
NFR34	Identify child-facing safeguards and the simulated publication steps.
Section 7	Plan the data entities, saving, retention, and deletion rules.

Phase 2: Proof of concept

Investigate whether the major project risks can be addressed through the independent experiments listed in Table 14. The linked requirements indicate what each experiment investigates rather than features that have been fully implemented. Privacy and offline safeguards shall be maintained throughout all experiments. Evaluate the results against the corresponding SRS requirements and project timeline. Approaches that do not demonstrate feasibility will be revised and tested again. By the December 4 PoC demonstration, the team will present the experimental evidence, design decisions, and the status of each identified risk. Risks will be classified as resolved, reduced, or still open, with any remaining work assigned to subsequent development phases.

Table 14: Phase 2: PoC experiments and investigated requirements

Risk and linked requirements	Experiment and expected evidence
Clinical content FR6, FR8, FR9, FR10, NFR2, NFR8, NFR32	Prepare a short eye-drop script and storyboard for clinician and child-life review. Record corrections, review rounds, and turnaround time. Seek a sample with no outstanding corrections and use the review effort to assess the feasible procedure scope.
Child interaction FR12, FR13, FR14, NFR3, NFR5, NFR9, NFR10	Try one narrated, visually guided interaction, such as picture matching. Record task completion, time, and assistance. Expert and adult feedback is preliminary; child testing requires consent and ethics approval. Retain child usability as an open risk if target-user evidence is unavailable.
Offline media NFR11, NFR12, NFR13, NFR14, NFR19, NFR21, NFR24	Run a standalone narration, animation, and touch sample in airplane mode on accessible devices. Seek iOS, Android, and older hardware where available. Measure startup, response, frame rate, synchronization, and size; estimate full media size and record untested configurations.

Phase 3: Core application

Use the PoC findings to build setup and local storage first, then the preparation journey, guidance, recovery, completion rewards, and caregiver controls. Table 15 lists the planned implementation for each requirement.

Content is reviewed before demonstration or release, while age-specific content is used where available. The target for this phase is a working core journey with its supporting quality controls, refined to the course scope.

Table 15: Phase 3: Planned core implementation

Requirement	Planned implementation
FR1	Show the caregiver notice before first use and retain access through the guide.
FR2	Store the acknowledgement locally and use it on later launches.
FR3	Show the planned procedure levels, illustrations, and progress states.
FR4	Allow any level to start without progress or reward restrictions.
FR5	Allow replay while keeping a completed level's status.
FR6	Develop the planned vision-testing, eye-drop, and slit-lamp preparation content.
FR7	Select age-specific content where the level provides it.
FR8	Explain possible sensations honestly before showing them.
FR9	Explain through narration and pictures that real procedures may differ.
FR10	Tell children they can express worry or discomfort to the doctor.
FR11	Display a clear ending and record the completed level.
FR12	Provide narrated instructions with visual demonstrations.
FR13	Offer encouraging hints and another attempt after a mistake.
FR14	Keep guidance visible and protect cue settings from accidental changes.
FR15	Retain visual guidance when sound is unavailable and show a caregiver prompt.
FR16	Reveal every correct answer promptly with narration and non-colour cues.
FR17	Validate questions at build time and exclude those without a correct answer.
FR18	Allow exercises to be skipped without penalties or blocked completion.
FR19	Accept one navigation input per transition and protect procedure narration.
FR20	Return to the journey map when the child selects the home control.
FR21	Stop the exited level's audio, animation, and visual effects.
FR22	Keep progress and rewards unchanged when a level ends early.
FR23	Start unfinished levels from the beginning after exit or closure.
FR24	Provide restart and exit controls while preserving saved progress.
FR25	Pause media during interruptions and resume when the interruption ends.
FR26	Save state changes locally and retain valid data if a save fails.
FR27	Recover from missing or invalid saves using new-installation data.
FR28	Record first-completion rewards and show what was earned.
FR29	Congratulate replays without issuing the same reward again.
FR37	Keep purchases limited to earned virtual currency.
FR38	Place caregiver settings behind an adult gate.
FR39	Provide caregiver explanations of the procedures and matching levels.
FR40	Support an optional nickname and neutral wording when it is absent.
FR41	Apply caregiver changes to age, audio, display, and accessibility preferences.
FR42	Confirm reset; clear progress and customization but retain notice acknowledgement.
FR43	Confirm deletion of all local data, including notice acknowledgement.
NFR1	Use consistent control positions across the implemented screens.
NFR2	Review content for age suitability and resolve frightening or misleading material.
NFR3	Size interactive elements for the target users on accessible devices.
NFR4	Keep a one-touch return control on activity screens.
NFR5	Design core tasks for use without reading or prior training.

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Table 15 – continued from previous page

Requirement	Planned implementation
NFR6	Make each level easy for a clinician to find from launch.
NFR7	Exclude streaks, expiring rewards, randomized rewards, and notifications from the reward design.
NFR8	Use short, familiar narration and readable caregiver text.
NFR9	Apply the accessibility criteria relevant to the implemented mobile interface.
NFR10	Distinguish states and prompts through cues beyond colour.
NFR11	Measure and improve startup time on accessible devices.
NFR12	Provide prompt visible responses to touch.
NFR13	Profile and improve animation frame rate on accessible devices.
NFR14	Synchronize narration with its accompanying animation.
NFR15	Keep effects and music within the narration's peak audio level.
NFR16	Check animations for flashing above the limit.
NFR17	Address crashes and freezes as the core workflows are integrated.
NFR18	Measure device temperature during sustained use where measurement is possible.
NFR19	Package the content needed for the implemented functions to work offline.
NFR20	Provide a development-installation path without accounts or extra content downloads.
NFR21	Measure installed size locally and optimize media; store-reported evidence is deferred.
NFR22	Record clinician approval of each level and guide entry before demonstration or release.
NFR23	Build reusable level and content components and record change effort.
NFR24	Adapt and test the application on devices and OS versions the team can access.
NFR25	Fix dependency versions and test changes before including them.
NFR26	Use the platform's private application storage.
NFR28	Prevent outbound application and library communications.
NFR29	Limit stored information to the application's defined local data.
NFR30	Check dependencies for known high- or critical-severity vulnerabilities.
NFR31	Review language, images, and character content for inclusive representation.
NFR32	Keep application content and draft listings free of medical claims.
NFR33	Record redistribution licences and required notices for included materials.
NFR34	Apply child-facing safeguards; plan the simulated publication workflow for Phase 6.

Phase 4: Feature integration

Add the engagement features in Table 16 to the working core for the Revision 0 demonstration. Their scale and visual detail may be refined to the remaining time and feedback. Recheck saving, reward consistency, reset behaviour, and inclusivity as the new features are integrated.

Table 16: Phase 4: Planned feature integration

Requirement	Planned implementation or integration
FR30	Show personal level progress, points, and coins without user comparisons.
FR31	Allow character selection without changing progress or rewards.
FR32	Develop a diverse character and accessory library within the available scope.
FR33	Show accessory previews, prices, and ownership.
FR34	Exchange earned coins for an unowned, affordable accessory.
FR35	Explain refused purchases without changing coins or ownership.

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Table 16 – continued from previous page

Requirement	Planned implementation or integration
FR36	Support owned accessories with one equipped item per category.
FR26, FR42, FR43, NFR31	Extend saving, reset, deletion, and inclusive review to the added customization data.

Phase 5: System verification

Evaluate the completed requirements and their corresponding fit criteria according to the V&V Plan, focusing on the areas outlined in Table 17. Use Table 6 to link the testing results to the stakeholder requirements. Testing with children will require appropriate consent and ethics approval. Testing with adults alone will not be sufficient to demonstrate that the application is usable by children or that it reduces their anxiety.

Table 17: Phase 5: Requirements revisited for verification

Requirements	Verification focus
Implemented functional requirements	Exercise the implemented use cases, including completion, replay, purchases, restart, save failure, and deletion.
NFR1, NFR3, NFR4, NFR5, NFR6, NFR8, NFR9, NFR10	Review usability and accessibility. Record task results and participant limitations; arrange approved child evaluation where possible.
NFR2, NFR7, NFR15, NFR16, NFR22, NFR31, NFR32	Review clinical accuracy, language, imagery, audio levels, and inclusion with the appropriate reviewers.
NFR11, NFR12, NFR13, NFR14, NFR17, NFR18	Measure performance and reliability on accessible devices, recording any unavailable measurements.
NFR19, NFR20, NFR21, NFR24	Test offline use, local installation, and size on the available test set; mark store-dependent or unavailable-device evidence separately.
NFR23, NFR25, NFR26, NFR28, NFR29, NFR30	Review change effort, private storage, retained data, network behaviour, and dependency checks.
NFR27, NFR33, NFR34	Review signing, licence records, and policy checks; identify simulated or outstanding publication evidence.

Phase 6: Final delivery

Address priority findings and rerun affected checks for the final demonstration. Table 18 covers delivery evidence, including the simulated Apple App Store and Google Play workflow. Real submission may be attempted if time and account access permit, with lower priority than the core application. Finalize source code, user guidance, traceability, test evidence, and limitations for Expo.

Table 18: Phase 6: Delivery requirements and course evidence

Requirement	Planned delivery evidence
NFR20	Demonstrate development-build installation and a mock store installation flow; attempt actual store installation if time permits.
NFR21	Report locally measured size; store-reported size remains pending actual publication.
NFR27	Check development-build signing and simulate official distribution steps; record any real submission separately.
NFR32	Review application notices and draft store listings for the educational purpose.

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Table 18 – continued from previous page

Requirement	Planned delivery evidence
NFR33	Finalize the licence log and required notices.
NFR34	Demonstrate mock listings and the submission checklist; attempt real submission as a lower-priority task and report actual approval status.

Course scope and future work. Table 19 outlines the limitations of the course project and possible future improvements, along with their related requirements and likely changes. For future improvements, the referenced functional requirements indicate which existing application features could be expanded.

Table 19: Course scope, simulations, and future work

Scope item	Related requirements and likely changes	Course treatment and future work
Store distribution	NFR20, NFR21, NFR27, NFR34, C8	The team will simulate the app store listing and submission process during the course. Actual publication will be attempted if time permits, but completing the application will take priority. Simulated submissions will be clearly distinguished from official app store approval. Installation testing, application size measurements, digital signing, and child safety requirements will still be addressed during the course.
Device coverage	NFR24, C1, LC9	The team will develop and test the application using the devices, operating system versions, and screen sizes available during the course. Testing results from physical devices and emulators will be reported separately. Any devices or configurations that could not be tested will be identified, with additional compatibility testing left for future work.
Augmented reality	LC12; related content: FR6	Augmented reality features will not be developed during the course. The team will focus on implementing the existing preparation games without augmented reality. Adding AR simulations to provide a more immersive experience may be considered in future development and would require additional design and testing.
Appointment-based personalization	LC13; related features: FR3, FR7, FR41	Personalizing preparation activities based on the procedures expected during a child's appointment will be left for future development. During the course, all preparation games will remain accessible to every child. The application will still support local caregiver preferences and age-appropriate content where available.

Any proposed changes to the project scope will be discussed with the supervisor. The team will then update the relevant requirements, development phases, and V&V Plan accordingly. The final deliverables will clearly identify features that were partially implemented, simulated, or postponed for future development.

13.3 Appendix C: Values of Auxiliary Constants

This appendix lists the auxiliary constants used throughout the requirements specification. These constants define quantitative thresholds, performance targets, platform compatibility requirements, and other fixed values referenced by the requirements. The Rationale column gives the source of each value. A value marked as an initial estimate has no external source yet and is to be confirmed as stated. Centralizing these values provides a single reference point for interpreting and verifying the specified requirements.

Table 20: Values of Auxiliary Constants

Constant	Value	Unit	Rationale
ADULT_TEXT_MAX_GRADE	8	reading grade	Caregivers vary in education and English proficiency (Section 2.4). Initial estimate, to be confirmed with caregivers.
ANSWER_REVEAL_MAX_S	1	s	Short enough that the child links the answer to the question. Initial estimate.
AV_SYNC_MAX_MS	100	ms	The same as RESPONSE_MAX_MS, the longest delay that feels instant [13]. Checked in the proof of concept.
CLINICIAN_FIND_MAX_MIN	1	min	From the time clinicians have per patient (Section 2.4).
CLINICIAN_TRAINING_MAX_MIN	5	min	From the time clinicians have for training (Section 2.4).
CONTROL_POSITION_TOLERANCE_MM	5	mm	One third of TOUCH_TARGET_MIN, so a control stays mostly over its usual position. Initial estimate.
CPU_UTILIZATION_MAX_PERCENT	80	%	Initial estimate to limit sustained CPU load on low-spec devices. To be validated through proof-of-concept testing alongside device temperature measurements.
CORE_TASK_MAX_MIN	5	min	Short enough for the attention span of a young child (Section 2.4). Initial estimate, to be confirmed in usability testing.
CRASH_FREE_RATE_MIN	99	%	With TEST_SESSIONS_MIN sessions, at most one session may fail. Initial estimate.
FLASH_MAX_PER_S	3	flashes	WCAG 2.2 success criterion 2.3.1 [6].
FRAME_RATE_COMPLIANCE_MIN	95	% of measured time	Allows brief drops, such as while a level loads. Initial estimate.

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Table 20 – continued from previous page

Constant	Value	Unit	Rationale
FRAME_RATE_MIN	30	frames per second	A common minimum for smooth animation on mobile devices. Initial estimate, measured in the proof of concept.
HEAT_TEST_SESSION_MIN	30	min	Longer than a typical session in a waiting room. Initial estimate.
INSTALL_SIZE_MAX_MB	300	MB	Small enough for older devices with little free storage (NFR21). Initial estimate, measured in the proof of concept.
LOW_SPEC_AGE_YEARS	5	years	Families may keep a phone or tablet for several years. Initial estimate.
MAINT_FRACTION	0.10	fraction of original development time	The example value of the course template [1]. A new level should cost far less than the first.
MIN_ANDROID	10	version	Initial estimate of a version that covers most devices still in use while keeping the test set small. To be checked against platform usage data.
MIN_IOS	16	version	Initial estimate of a version that covers most devices still in use while keeping the test set small. To be checked against platform usage data.
RESPONSE_MAX_MS	100	ms	The longest delay that feels instant to a user [13].
RESPONSE_RATE_MIN	95	%	Allows occasional delays, such as while a level loads. Initial estimate.
SCREEN_MAX_IN	13	inches	The largest current tablet screens, such as the 13-inch iPad Pro.
SCREEN_MIN_IN	4.7	inches	The smallest current phone screens, such as the 4.7-inch iPhone SE.
SENTENCE_MAX_WORDS	10	words	Initial estimate, to be confirmed with the child-life specialists (NFR8).
SKIP_HOLD_MIN_S	1	s	Long enough that a repeated tap does not skip, and short enough for a child to skip on purpose. Initial estimate, to be confirmed in usability testing.
STARTUP_MAX_S	5	s	Initial estimate of how long a young child waits before losing interest. Revisited after the proof of concept.

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Table 20 – continued from previous page

Constant	Value	Unit	Rationale
STARTUP_TRIALS_MIN	10	cold-start trials	Enough trials for a stable median.
TEST_SESSIONS_MIN	100	sessions	Enough sessions to measure CRASH_FREE_RATE_MIN.
TOUCH_TARGET_MIN	15	mm	About twice the 9 mm minimum of Material Design [12], for less precise motor control (NFR3). To be confirmed in usability testing.
UNAIDED_RATE_MIN	80	%	Initial estimate, to be confirmed in usability testing.

13.4 Appendix D: Generative AI Disclosure

AI Tools Used

- Claude (Opus 5.5)
- ChatGPT (GPT-6 Luna, GPT-6.1 Sol)

How and Where Used

ChatGPT and Claude were used as supporting tools during the preparation of this document, primarily to assist with LaTeX formatting, improving wording and clarity, and organizing information. They were used in the following sections:

- In the sections listed below, the ideas and content were developed by the team. Opus 5.5 was used to improve wording, grammar, and flow while preserving the team's original meaning and decisions.
  - Sections 1.1, 1.2, and 2.1
- In Section 8 (Non-Functional Requirements), Opus 5.5 was provided with stakeholder meeting notes, team assumptions, and preliminary requirements to generate an initial draft of the NFRs. The team then manually reviewed, refined, and validated the generated requirements. Opus 5.5 was also used to identify which functional requirements and which safety requirements each non-functional requirement traces to.
- In Section 10 (Rationale), Opus 5.5 was used to identify connections and shared justifications across requirements, which the team then used to develop the section.
- In Section 2.2 (Product Vision), ChatGPT was used to help summarize and organize the application's purpose and goals based on the team's existing Problem Statement and Goals document. The wording was refined to ensure the product vision remained concise, understandable, and consistent with the project's objectives.
- In Section 7 (Data Requirements), ChatGPT was used primarily to assist with formatting data dictionary tables in LaTeX, organizing major data entities and their attributes, and improving the readability of the Privacy and Data Retention subsection.
- In Appendix B (Section 13.2), ChatGPT was used to organize the existing requirements into implementation phases using the Development Plan milestones and to draft the explanations, priority justifications, and linked requirement tables in LaTeX.
- Opus 5.5 was used to identify constant values in Section 8 (Non-Functional Requirements), generate Table 20, and establish links between the constants.
- In Section 2.5 (System Constraints), ChatGPT was used to improve the clarity and conciseness of the system constraints written by the team member who authored the section. It also provided feedback on whether any of the constraints were too vague to be valid system constraints.
- In Section 4 (User Personas), ChatGPT was used to review and improve the personas written by the team member who authored the section. Specifically, ChatGPT was prompted to compare the initial draft against the VARIED (Vivid, Actionable, Real, Identifiable, Exact, Detailed) criteria from SFWRENG 3RA3 and provide feedback for the team member to integrate. Once the content of the personas was solidified, ChatGPT was used to refine the wording for a more professional presentation.
- In Section 2.3 (System Context), Opus 5.5 was used to generate the system context diagram and to assist with LaTeX formatting.
- In Section 6 (Functional Requirements), Opus 5.5 was used for LaTeX formatting and for linking the requirements for traceability.

- In Section 6.1 (Formal Model), Opus 5.5 was used to help brainstorm ideas for the formal model. The team selected which ideas to keep. Opus 5.5 was then used to format the selected ideas in LaTeX. This included the formatting of the equations, definitions, and theorems. Opus 5.5 also helped provide traceability between the formal model and the functional requirements.
- In Section 9 (Requirements Traceability Matrices), Opus 5.5 was used to generate the traceability tables and to link the requirements together.
- In Section 1.3 and Section 2.4, Opus 5.5 was used to brainstorm the descriptions of stakeholders and their characteristics using previous stakeholder definitions from Problem Statement and Goals document. The included descriptions drew on these results, with heavy editing and review of the AI-generated content.
- In Section 3, Opus 5.5 was provided with stakeholder meeting notes, preliminary requirements, the project description and problem statement to generate an initial draft of stakeholder requirements. The team then manually reviewed, refined, excluded, and validated the generated requirements.
- In Section 9, Opus 5.5 was used to generate the traceability between stakeholder requirements and use cases. These traceability connections were thoroughly checked one at a time by the team to ensure alignment between the two.

Verification of AI-Generated Content

The team reviewed AI-assisted material to ensure it was consistent with the project's scope, existing documentation, and intended application functionality. The team reviewed suggested wording, data entities, and recommendation approaches before including them in the document. Initial design ideas were also noted as needing further feedback from the project supervisors.

AI-assisted LaTeX formatting was checked to ensure that tables and other document elements were presented correctly. The team remains responsible for the accuracy, completeness, consistency, and appropriateness of the final SRS document.

13.5 Appendix E: Reflection

Questions 1 and 2 were answered individually, while Questions 3 and 4 were answered as a team. Questions 5 and 6 contain both team and individual components. The team responses are provided under Team Reflection, while the individual responses are provided under each respective team member's name.

Team Reflection

3. **How many requirements were inspired by client interactions or their proxies?**

   Approximately 30 requirements were inspired by discussions with our clients, stakeholders, peers, or potential users. These interactions primarily informed our stakeholder requirements and the functional requirements derived from them, helping us understand the application's goals, intended users, and valuable features for preparing children for paediatric ophthalmology appointments.

   However, we did not have sufficient time to discuss every requirement with stakeholders individually. We had approximately two weeks to complete the SRS, so we prioritized discussions about major decisions that would significantly affect the system if changed later.

   Consequently, many non-functional requirements were developed through team analysis, research, technical considerations, and assumptions about children aged 3–7 and their parents. In particular, our measurable usability and performance targets will need further validation. In future iterations, we would seek additional stakeholder feedback to verify these assumptions and refine the requirements.

4. **Which courses have helped or will help your team succeed with the capstone project?**

   Several courses have provided a foundation for this project. SFWRENG 3RA3, which focused on software requirements, was particularly relevant because we previously developed a Software Requirements Specification and learned about requirements elicitation, documentation, and traceability. This experience made the structure and purpose of the current deliverable more familiar.

   SFWRENG 3A04, which focused on large system design, was also useful because it introduced concepts and activities similar to those involved in a capstone project. Designing a larger software system helped us understand the importance of defining scope, identifying system constraints, and considering interactions between system components.

   Additionally, SFWRENG 4HC3 was helpful because it involved interviewing stakeholders and eliciting requirements. These skills helped us identify important questions to ask our stakeholders and distinguish their needs from our own assumptions about the application.

5. **What knowledge and skills will the team collectively need to acquire?**

   The team identified five knowledge areas essential to completing the capstone project:

   - **Paediatric ophthalmology and clinical procedures:** We need to understand common paediatric eye examinations, including their purpose, steps, and what children may experience. This will help us represent procedures accurately and prepare children for their appointments.
   - **Child development and child-friendly design:** We need to understand the cognitive, emotional, and motor capabilities of children aged 3–7. This will inform decisions about language, instructions, visual presentation, and navigation to ensure the interface is appropriate for the target age group.
   - **Mobile application development:** We need the technical skills to implement navigation, interactive components, animations, audio, and local data storage. We must also ensure compatibility across supported platforms and devices.

- **Accessibility and usability testing:** We need to evaluate whether children and parents can use the application effectively. This includes assessing touch targets, text readability, colour contrast, navigation, and alternative ways of presenting information to identify and address usability barriers.
- **Educational game design:** We need to design interactive activities that reinforce learning objectives and familiarize children with ophthalmology procedures. This includes selecting appropriate game mechanics, feedback, and rewards that support learning while maintaining engagement.

These areas cover the clinical, developmental, technical, and educational needs of the project. Each team member will select at least one area based on their experience, responsibilities, and learning goals, and describe their plans for developing the relevant knowledge or skills.

6. **What are at least two approaches to acquiring the knowledge or mastering each skill?**

   The team identified two approaches for each knowledge area.

   - **Paediatric ophthalmology and clinical procedures:**
     (a) **Review clinical resources.** We will study reliable medical resources and educational materials to understand common paediatric eye examinations and their procedures.
     (b) **Consult clinical experts.** We will ask clinicians and our project supervisor to clarify procedural details and review our understanding, helping us identify inaccuracies.
   - **Child development and child-friendly design:**
     (a) **Study child development resources.** We will review research on the cognitive, emotional, and motor development of children aged 3–7 to inform decisions about language, instructions, visuals, and interactions.
     (b) **Develop and review interface prototypes.** We will create screen mock-ups and navigation prototypes and review them with our supervisor and relevant stakeholders to identify confusing layouts, unclear instructions, and developmentally inappropriate interactions.
   - **Mobile application development:**
     (a) **Follow official documentation and tutorials.** We will use the selected development framework's documentation and tutorials to learn how to implement navigation, interactive components, animations, audio, and local data storage.
     (b) **Build small technical prototypes.** We will implement individual features before integrating them into the application to practise using the framework and identify technical, integration, and compatibility issues.
   - **Accessibility and usability testing:**
     (a) **Study accessibility guidelines.** We will study WCAG 2.2 and relevant mobile accessibility guidance to evaluate text readability, colour contrast, touch targets, navigation, and alternative ways of presenting information. We will use checklists and evaluation tools to identify accessibility issues.
     (b) **Conduct structured usability evaluations.** We will define tasks such as navigating to a procedure, completing a walkthrough, and using a practice activity. We will record difficulties, evaluate task completion, and use the findings to prioritize improvements. Where appropriate and approved, we will also gather feedback from representative users or stakeholders.
   - **Educational game design:**

(a) **Study educational game design principles.** We will review resources on game-based learning, educational objectives, feedback mechanisms, and age-appropriate game mechanics to design activities that reinforce understanding of ophthalmology procedures.

(b) **Prototype and evaluate gameplay.** We will develop small gameplay prototypes and assess whether their mechanics, feedback, and activity sequences support the intended learning objectives. Feedback from our supervisor and relevant stakeholders will help us identify ineffective mechanics and improve the connection between gameplay and learning.

Team members will select approaches that align with their learning goals and describe their plans in their individual reflections.

Mohammad Bilal

1. **What went well while writing this deliverable?**

   One aspect that went well was that we established a clear understanding of the application's purpose, scope, and features that would be excluded from the initial release. This provided a strong foundation for developing the requirements and helped the team remain aligned on the project's overall direction. Our project supervisor was also responsive to our questions, allowing us to clarify uncertainties and make decisions without significant delays.

   Another factor that contributed to our progress was dividing the workload among team members early in the process. Assigning clear responsibilities helped us stay organized and manage the limited time available to complete the deliverable. From my perspective, having a well-defined scope and clear responsibilities made it easier to prioritize my assigned sections, work efficiently, and complete them on schedule.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   One pain point was duplication and inconsistency across the requirements. Since multiple team members worked on different sections in parallel, similar ideas sometimes appeared in multiple places, and an idea described as an assumption in one section could be presented as a requirement in another. Resolving these inconsistencies was challenging because changes in one section could affect other parts of the SRS. We addressed this by keeping pull requests relatively small and having the entire team review changes that could introduce conflicting or duplicate information. This helped us identify inconsistencies, clarify the intended meaning, and remain aligned as the document evolved.

   Another challenge was the limited timeline. We had approximately two weeks to complete the SRS, compared with an entire academic term for a similar process in a previous requirements engineering course. Conducting enough stakeholder interviews to validate every requirement was therefore not feasible. I addressed this by prioritizing consultation on major decisions and making reasonable assumptions about smaller details when necessary. Although these assumptions helped us progress, they will need to be revisited as we receive further feedback and develop a stronger understanding of the project's domain.

5. **Which knowledge areas or skills will you personally need to acquire?**

   I will focus on accessibility and usability testing. Since the application is intended for children aged 3–7, it must accommodate their developing motor skills, limited reading ability, and varying accessibility needs. I need to learn how to evaluate whether the interface is intuitive, accessible, and easy to navigate without relying heavily on adult assistance. This includes assessing touch target sizes, text readability, colour-independent cues, navigation consistency, and the clarity of audio-visual instructions. Developing these skills will help me identify usability barriers and contribute to an application that is inclusive and appropriate for its intended users.

6. **What approaches will you pursue to acquire this knowledge or master this skill, and why?**

   I will focus on studying established accessibility guidelines. I will review WCAG 2.2 and relevant mobile accessibility guidelines to understand established practices for designing accessible interfaces. I chose this approach because the application is intended for children aged 3–7, who may have limited reading ability, developing motor skills, and varying accessibility needs. These guidelines will help me understand how to evaluate important aspects of the application, such as touch target sizes, text readability, colour contrast, and alternative ways of presenting information. By learning and applying these established practices, I will be better equipped to identify potential accessibility barriers and recommend improvements to the application's design. This approach will also provide me with a reliable foundation for making informed accessibility decisions throughout development, rather than relying solely on personal judgment or assumptions about what makes an interface easy to use.

Ahsan Muzammil

1. **What went well while writing this deliverable?**

   One thing that went well was being able to use our previous deliverables, such as the Problem Statement and Goals and Development Plan, as a foundation for the SRS. This helped me better understand the purpose of the application and how the different sections of the SRS should relate to one another. I also gained a better understanding of how requirements influence other parts of the document, particularly the use case scenarios, data requirements, and development plan. Working on these sections helped me think more carefully about how the application is expected to function and how to document these expectations clearly.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   One of the main challenges was maintaining consistency across the document and avoiding duplicated or contradictory information. Since different team members were responsible for different sections, I had to regularly compare my work with other sections to ensure that the terminology, data entities, and described functionality remained consistent.

   Another major challenge was the limited timeline. We only had approximately two weeks to develop the SRS, which was not enough time to confirm every requirement and design decision with our supervisors. Since our supervisors are busy medical professionals, our meetings mainly focused on high-level decisions rather than every individual feature. As a result, we had to make assumptions and document some requirements based on functionality we anticipated implementing, even though we had not yet confirmed their feasibility or received approval for every detail. We addressed this by using our existing project goals to guide these decisions and identifying areas that would require further clarification.

   I also experienced difficulties with dependencies between sections. For example, I was responsible for Section 5 (Use Case Scenarios) and Appendix B (Development Plan), but I could not fully complete these sections until the functional and non-functional requirements were sufficiently developed. The use cases also depended on the user personas in Section 4. This made it difficult to work at my own pace and resulted in some sections being completed close to the deadline. I addressed this by working on the portions that did not depend on other sections first, then reviewing and updating the remaining content as my teammates completed their work.

5. **Which knowledge areas or skills will you personally need to acquire?**

   One of the main skills I would like to develop is mobile application development. Although I have previous programming experience, I have limited experience developing mobile applications, particularly those involving interactive interfaces, animations, audio, and local data storage.

I will need to learn how to use our selected development framework, Unity, and C# to implement these features. I also want to understand how to structure a mobile application, manage navigation between screens, handle user interactions, and ensure the application functions correctly across supported devices. Developing these skills will help me contribute to the implementation of EyeCanDo.

6. **What approaches will you pursue to acquire this knowledge or master this skill, and why?**

   The approaches I plan to pursue to acquire knowledge about mobile application development include the following:

   First, I will use official documentation and tutorials to learn the fundamentals of Unity and C#, including interface development, navigation, animations, audio, and local data storage. This will help me understand the development framework and become familiar with the tools needed to implement our application.

   Second, I will build small technical prototypes to practice implementing individual features, such as navigating between screens, handling user interactions, and saving progress locally. Developing these features separately will allow me to apply what I have learned, identify technical and compatibility issues, and gain experience before integrating them into EyeCanDo.

Jackson Beach

1. **What went well while writing this deliverable?**

   Although there was a large amount of work involved in this deliverable, I found that our team was able to split it up well into manageable pieces. Our reviews of each other's work were timely and provided actionable feedback, which helped improve the overall quality of the document. I also enjoyed working on the Personas section, as it allowed me to step into the proverbial shoes of our target users and consider their needs and goals when interacting with the application. Additionally, the nature of our application made the system constraints easier to identify, since we need to be careful about what data we collect and how it is stored or transmitted.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   The main pain point was the limited time we had to work on the SRS and Hazard Analysis, which meant we were not able to fully elicit requirements from our stakeholders. We worked around this by using the interactions we did have, including emails with our supervisor, Daiana, and our team liaison, Ethan, visiting the urgent care clinic to get firsthand experience. These interactions helped us make educated assumptions where we still had gaps in our understanding. As the document grew, it also became more difficult to maintain consistency and avoid redundancy. We managed this by keeping active discussions going on Discord and reviewing and merging pull requests promptly.

5. **Which knowledge areas or skills will you personally need to acquire?**

   The main area I need to learn more about is the domain of our application, specifically the different preparation activities that take place before a paediatric ophthalmology appointment and the sensations a patient might experience during them. In terms of technical skills, I need to become more familiar with Unity, since it is the game engine we plan to use and I have not used it before. I will also need to learn how to develop for mobile devices in Unity.

6. **What approaches will you pursue to acquire this knowledge or master this skill, and why?**

   To build my domain knowledge, I will continue asking our supervisor questions, both directly and through our team liaison. Once we have research ethics approval, being able to speak with patient families will also be helpful. They can describe what they have seen their children go through and the sensations or concerns they may have experienced, such as burning eye drops or uncertainty about

a piece of equipment. Direct conversation is the best approach for me here, since I have very little domain knowledge. Being able to speak with people who have this experience and ask questions as they come up will help me fill in the gaps.

For Unity development, I will use YouTube tutorials, AI tools, Unity documentation, and input from peers with more Unity game development experience than I have. I think using several sources will help me work through questions and fill in gaps as I learn throughout the development process.

Jake Finlay

1. **What went well while writing this deliverable?**

   Dividing the workload in advance through GitHub worked well again. Each section had an owner, so everyone knew what they were responsible for and could start without waiting to be assigned work. We had also discussed the scope of the project and our expectations for the application many times before starting. As a result, the team was already on the same page about what the requirements should cover.

   Our supervisor also responded to our questions relatively quickly. Their feedback turned out to be very important for this document. For example, they had the following recommendations: make every level available at all times and present the levels as a journey. This helped us shape the functional requirements, data requirements, and formal model.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   My main pain point was dependencies between sections. I was responsible for the functional requirements, which depended on stakeholder requirements and use cases. Those sections took a while to finish, so I had to wait before I could properly start. I already had a good picture of the requirements, so I was able to write most of them early. However, I still had to provide traceability for requirements that had not been written yet. I resolved this by writing my sections first and then revisiting them to update the traceability once the other sections were finished.

   Traceability with the Hazard Analysis was another pain point. Even after the safety and security requirements were finished, it was tedious to check that the references matched and that the links between the two documents worked. To make this easier, I added links that open the matching requirement or hazard directly in the other document. I also added a table that traces each safety and security requirement to the functional requirements that satisfy it.

   The last pain point was the timeline. In SFWRENG 3RA3, we had an entire term to prepare a requirements document. In this course, we had about two weeks. This project also has a real supervisor and real stakeholders. Our supervisor is busy, so we had few opportunities to elicit requirements. Ideally, we would have spent much longer on this stage. For example, we could have built cheap proofs of concept with the help of AI to show our supervisor what we had in mind. This would have given us a much clearer picture of the application and its requirements. Since we did not have this time, we made educated guesses where necessary so that we could meet the deadline. We will revisit this document once we receive more feedback from our supervisor.

5. **Which knowledge areas or skills will you personally need to acquire?**

   I will focus on educational game design. The purpose of EyeCanDo is to teach children about their procedures so that they are less nervous at their appointment. A game that is fun but teaches nothing would not meet this goal. A game that teaches but is not engaging would not be played long enough to help. I need to be able to evaluate both sides of this balance.

   First, I need to learn how to define what each level should teach. Each level should have clear learning objectives, such as the steps of the procedure, the equipment the child will see, and what the child

may feel. Once these objectives are clear, I need to learn how to choose game mechanics that support them. For example, practising holding still for the eye drops teaches more than a general mini-game with an eye theme.

Second, I need to learn how educational content is presented effectively in a digital format. This includes teaching strategies such as repetition, short steps, and combining narration with pictures. It also includes how feedback and rewards affect learning. Rewards should encourage the child to keep learning without distracting them from the content.

Finally, not every child learns the same way. Our users range from three to seven years old. Some will learn best by watching, while others will learn best by doing. Some children will also have different abilities and needs. I need to learn how to design activities that are useful for as many children as possible, rather than only for a typical child.

6. **What approaches will you pursue to acquire this knowledge or master this skill, and why?**

   I will pursue both approaches identified by the team.

   First, I will study educational game design principles. I will review resources on game-based learning, multimedia learning, and feedback in educational games. I will also look at existing games and apps that prepare children for medical visits, and at the medical play techniques used by child-life specialists. I chose this approach because these principles have already been tested by others. Learning them first will help me avoid common mistakes, such as rewards that pull attention away from the content or mechanics that do not relate to the learning objective.

   Second, I will prototype and evaluate gameplay. I will build small and cheap prototypes of possible level designs, using AI tools to speed this up where appropriate. I will then check each prototype against the learning objectives of its level and review it with our supervisor. Where approved, we will also gather feedback from representative users. I chose this approach because it addresses the main gap from this deliverable. We did not have time to show our supervisor concrete ideas, so some requirements are still educated guesses. Prototypes will give our supervisor something concrete to react to. They will also help us decide the format of each level, which is still open in the requirements.

Ethan McMehen

1. **What went well while writing this deliverable?** I found the organization and structure of my parts simplified my writing process. Since I had the stakeholder and user characteristics parts of the document, which I previously worked on in the Problem Statement and Goals, I had the organization to complete my 3 sections well. Writing the stakeholders and user characteristics sections, with inspiration from the previous document was straightforward. I also found that making the requirements was clear too.

   I found the structure of this deliverable also useful. Every requirement has a specific code that makes it easy to track and reference work throughout the document. Writing in tables also went well because of this, since any traceability matrices or other tables were clear to write using all of the acronyms for the requirements.

   I also found our team communication went well in this deliverable because we were able to effectively communicate our thoughts about the game from stakeholder information. We applied the results of these discussions from our Discord into the work, which kept our ideas and concepts aligned across the requirements sections.

2. **What pain points did you experience during this deliverable, and how did you resolve them?**

   The biggest pain point for me was traceability. This involves ensuring everything is connected whilst not repeating itself. Since each of us was taking a requirements section it is easy to duplicate requirements

or generate requirements classes. This meant that after writing the requirements, I first had to check if there were duplicates. Then I had to connect all the requirements together to ensure traceability. This took a very long time, a huge amount of focus, and constant communication with the team.

I solved this pain point by using Claude (Opus 5.5) to do the heavy lifting in performing the original traceability mappings. After receiving a list of the mappings, I cross-referenced the work against each other. In my specific instance, I had to map the stakeholder requirements to the use cases. To do this effectively, I used the Claude list and read through the use case and every requirement that connected to it to ensure there actually was a connection. After validating all of this work, I included it in the table for additional review later.

3. **Which knowledge areas or skills will you personally need to acquire?**

   I would like to acquire knowledge in two of the areas we described in the team answer. Firstly, knowledge in paediatric ophthalmology and clinical procedures. Secondly, in child development and child-friendly design for the video game. As the team liaison, it is my responsibility to hold communication between the stakeholders and the team. This means I will be a key communicator and organizer of client visits (with the supervisor), validation testing procedures with the primary users (children and caregivers), and discussions with other stakeholders who are experts in child development and design (child-life specialists and child development experts like McMaster University professors).

   This effectively means I will need to develop and maintain professional communication skills with the stakeholders, and then be able to communicate this information to the team. I will also have to learn the procedures and the knowledge scope of these professionals on a surface level to provide strong verification in reviews and testing amongst the team development.

4. **What approaches will you pursue to acquire this knowledge or master this skill, and why?** I will take two approaches to acquire the knowledge for our project. The first is constant communication. I will strive to maintain communication through the mediums we have learned about in 3RA3 and 4HC3. This includes emails with supervisors and clinicians, surveys and questionnaires for children, parents, and ophthalmologists, and in-person visits to the clinic to gain real life experience, connections, and knowledge of the experiences of our stakeholders. In this way, I will ensure that our team will get feedback and data to guide our development and improve the application to ensure we follow stakeholder requirements. All of this will be achievable too, once our team's study is approved by the Hamilton Integrated Research Ethics Board (HiREB), which will officially permit us to collect data and participate in direct communication methods with patients. Our team is currently working to achieve this ethics status.

   The second approach is research. To understand clinical procedures and examinations the children will experience, research through procedure guides or YouTube videos is my plan. I think this is an effective plan because our supervisor has already sent a YouTube video to learn about a procedure, which showed us that we can gain surface level knowledge from such sources. In conclusion, I think my communication techniques and research background can guide me to acquire the knowledge base needed while simultaneously developing my communication skills.

References

[1] W. Spencer Smith. capTemplate: Template for Capstone Projects. https://github.com/smiths/capTemplate, 2022. GitHub repository.

[2] ISO/IEC/IEEE. Systems and Software Engineering — Life Cycle Processes — Requirements Engineering. International Standard ISO/IEC/IEEE 29148:2018, 2018.

[3] ISO/IEC/IEEE. Systems and Software Engineering — System Life Cycle Processes. International Standard ISO/IEC/IEEE 15288:2023, 2023.

[4] Mayo Clinic College of Medicine and Science. Child Life Specialist. https://college.mayo.edu/academics/explore-health-care-careers/careers-a-z/child-life-specialist/, 2026. Accessed 2026-10-15.

[5] J. L. Sujuan, S. Handa, C. Perera, and A. Chia. The psychological impact of eyedrops administration in children. *Journal of American Association for Pediatric Ophthalmology and Strabismus*, 19(4):338–343, Aug 2015.

[6] World Wide Web Consortium. Web Content Accessibility Guidelines (WCAG) 2.2. W3C Recommendation, https://www.w3.org/TR/WCAG22/, 2023. Published 2023-10-05.

[7] Province of Ontario. Accessibility for Ontarians with Disabilities Act, 2005, S.O. 2005, c. 11. https://www.ontario.ca/laws/statute/05a11, 2005. Ontario legislation establishing accessibility standards for people with disabilities.

[8] Health Canada. Guidance Document: Software as a Medical Device (SaMD): Definition and Classification. https://www.canada.ca/en/health-canada/services/drugs-health-products/medical-devices/application-information/guidance-documents/software-medical-device-guidance-document.html, 2019. Adopted 2019-10-03; effective 2019-12-18.

[9] Apple Inc. App Store Review Guidelines. https://developer.apple.com/app-store/review/guidelines/, 2026. Accessed 2026-10-12.

[10] Google LLC. Google Play Developer Program Policies. https://play.google/developer-content-policy/, 2026. Accessed 2026-10-12.

[11] Apple Inc. Human Interface Guidelines. https://developer.apple.com/design/human-interface-guidelines/, 2026. Accessed 2026-10-12.

[12] Google LLC. Material Design 3. https://m3.material.io/, 2026. Accessed 2026-10-12.

[13] Jakob Nielsen. *Usability Engineering*. Morgan Kaufmann, Boston, MA, USA, 1993.